

Introduction of Computer Vision



Industrial Computer Vision Course
Spring 2026, Lecture 1
Lecturer: Youngbae Hwang

Syllabus

- Evaluation

- Attendance: 20%, Homework: 40%, Midterm/Final Project: 40%

- Course schedule

- 2nd Week - Input/Output & Graphic User Interface (GUI) (ch.1)
- 3rd Week – Histogram & Filtering (ch. 2)
- 4th Week – Frequency-based Image Processing (ch. 2)
- 5th Week – Boundary Extraction (ch. 3)
- 6th Week – Image Segmentation (ch. 3)
- 7th Week – Corner detection (ch. 7)
- 8th Week – Midterm Project
- 9th Week – Invariant Feature & Descriptor (ch. 7)
- 10th Week – Correspondence Search (ch. 8)
- 11th Week – Optical Flow & Panorama Imaging (ch. 8)
- 12th Week – Camera Calibration (ch. 9)
- 13th Week – Stereo Camera Calibration (ch. 9)
- 14th Week – Object Detector (ch. 4)
- 15th Week – Final Project

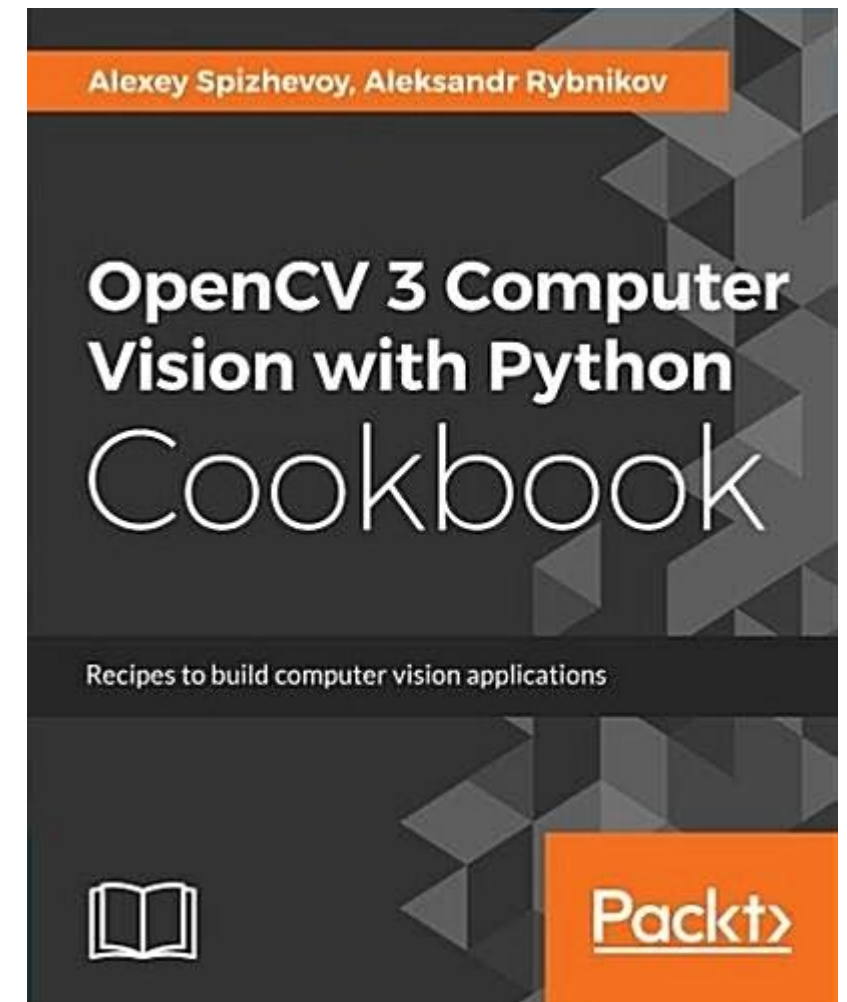
TextBook

- 교재

- 파이썬과 OpenCV를 이용한 컴퓨터 비전 학습
- OpenCV-Python으로 배우는 영상 처리 및 응용

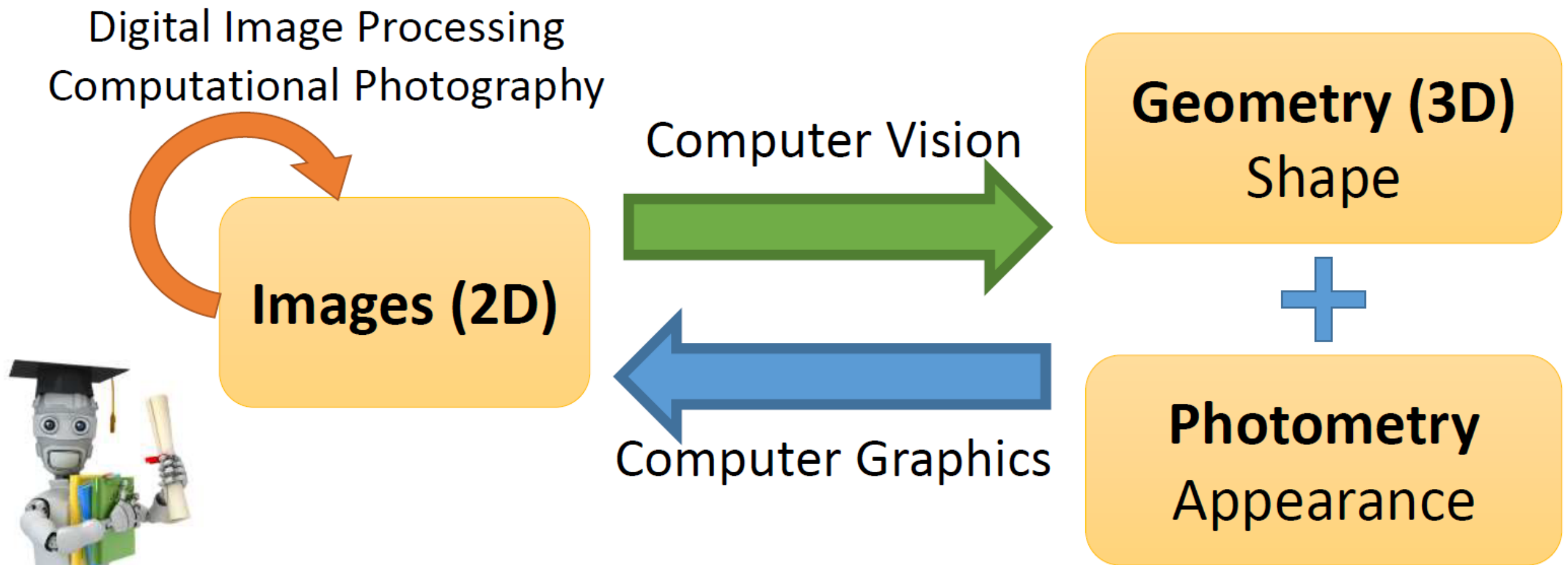
- Contents

- 1: I/O and GUI
- 2: Matrices, Colors, and Filters
- 3: Contours and Segmentation
- 4: Object Detection and Machine Learning
- 5: Deep Learning
- 6: Linear Algebra
- 7: Detectors and Descriptors
- 8: Image and Video Processing
- 9: Multiple View Geometry



What is
computer vision?

Computer Vision and Nearby Fields



Machine learning:

Vision = Machine learning applied to visual data

Computer Vision matters



Safety



Health



Security



AR/VR/MR

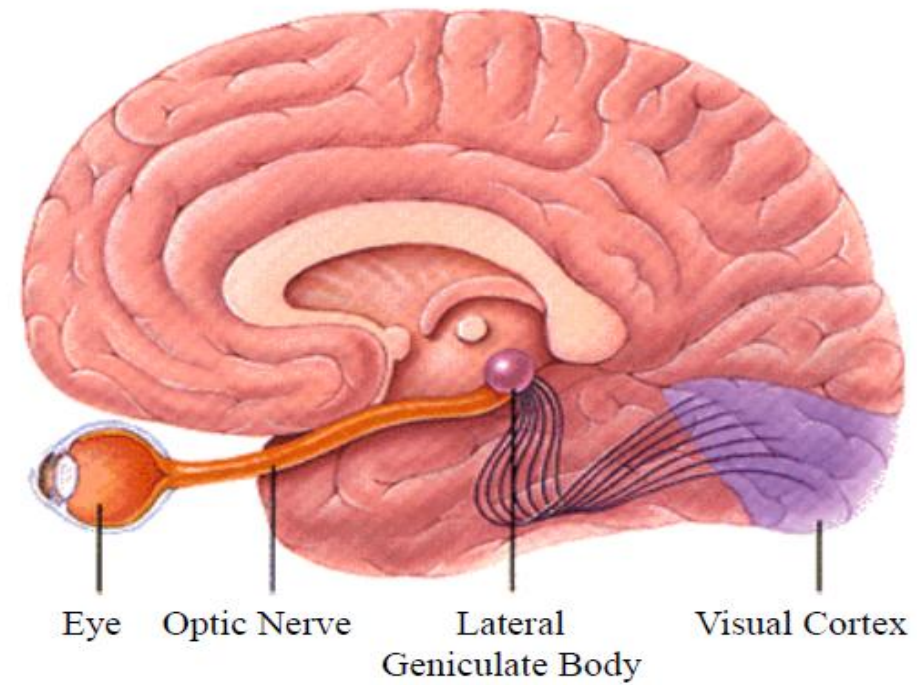
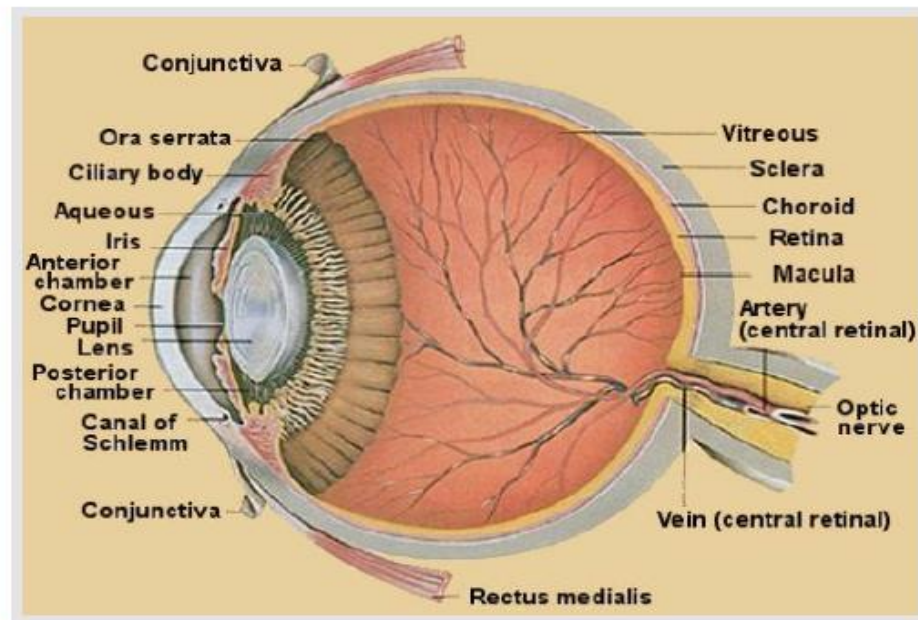


Movie



Robot

Human vs. Computer Vision





What a person sees

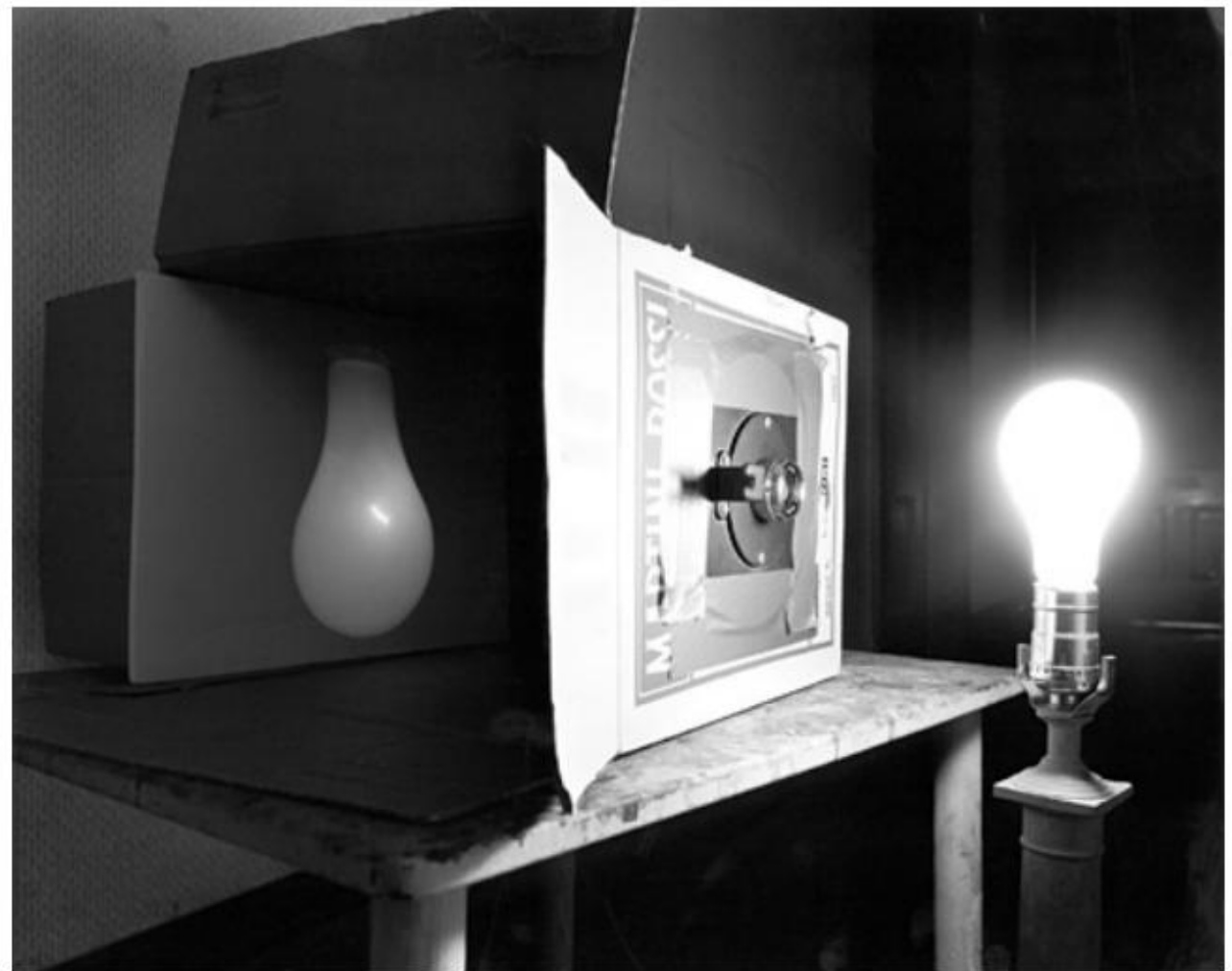


Why are we able to interpret this image?

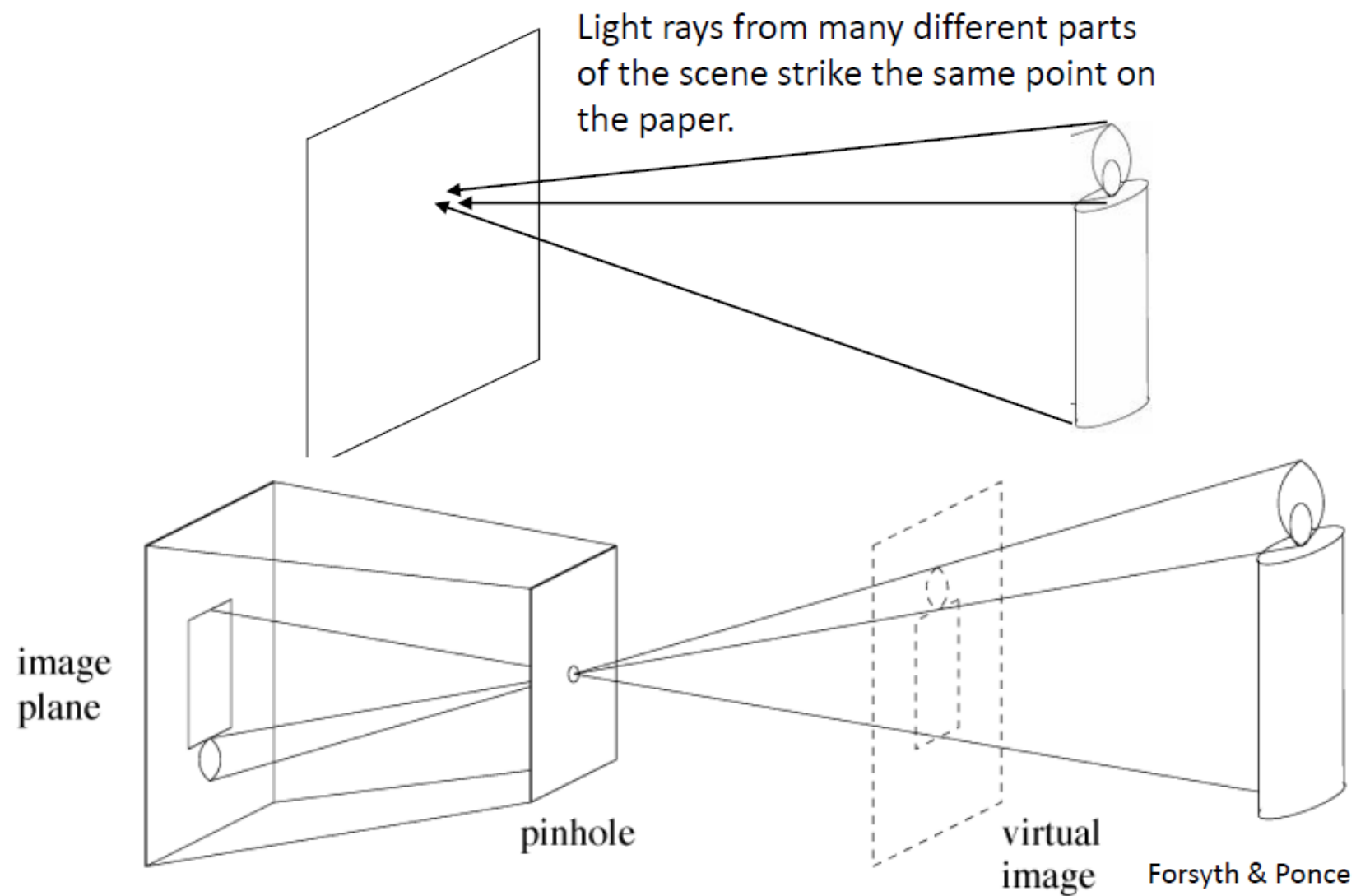
The goal of computer vision is
to give computers
(super) human-level perception

Image formation on the backplate of a photographic camera

A camera obscura device consists of a box, tent or room with a small hole in one side. Light from an external scene passes through the hole and strikes a surface inside, where the scene is reproduced, inverted (thus upside-down) and reversed (left to right), but with color and [perspective](#) preserved.



Pinhole camera model



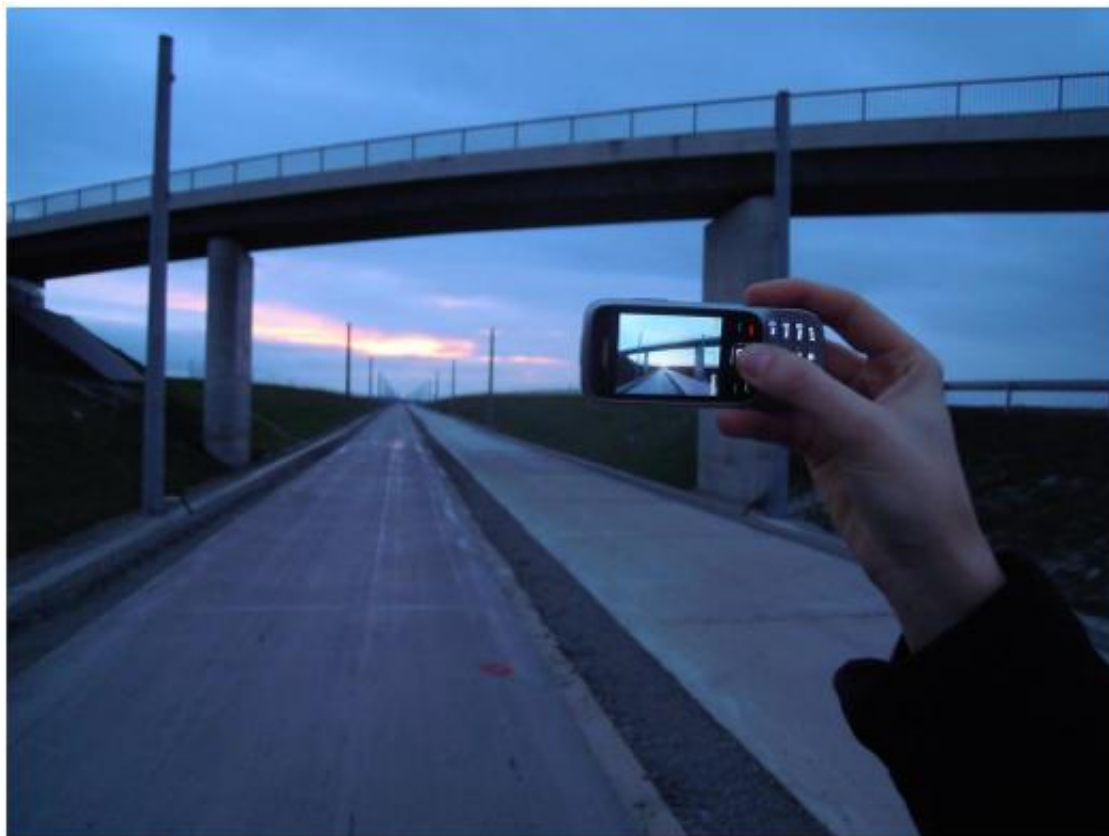
The pinhole camera only allows rays from one point in the scene to strike each point of the paper.

Computer Vision

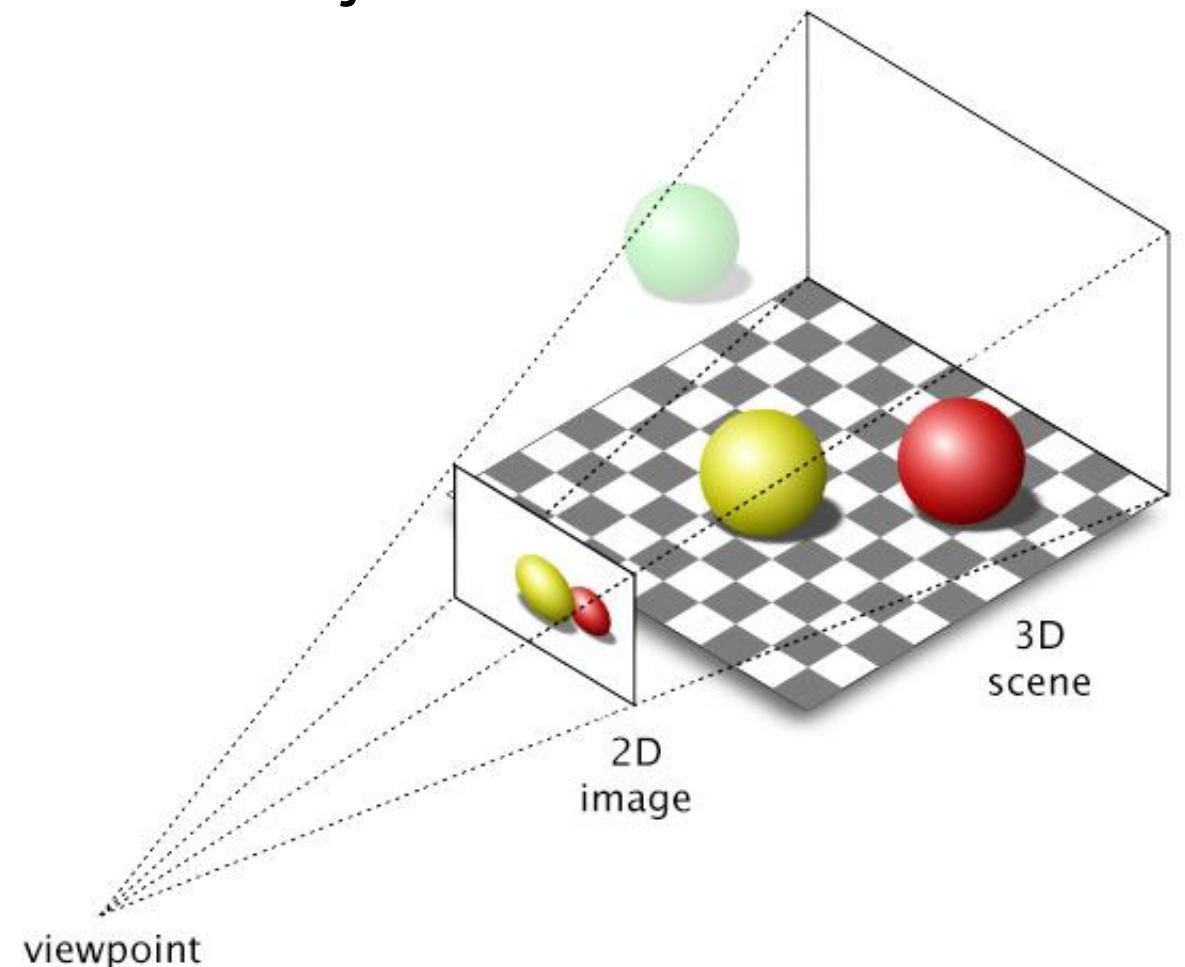
Forsyth & Ponce

What is an Image?

- Image : Projection of the 3D world onto an (2D) image plane
 - 2-dimensional patterns of brightness values
 - Formed by the projection of 3D objects

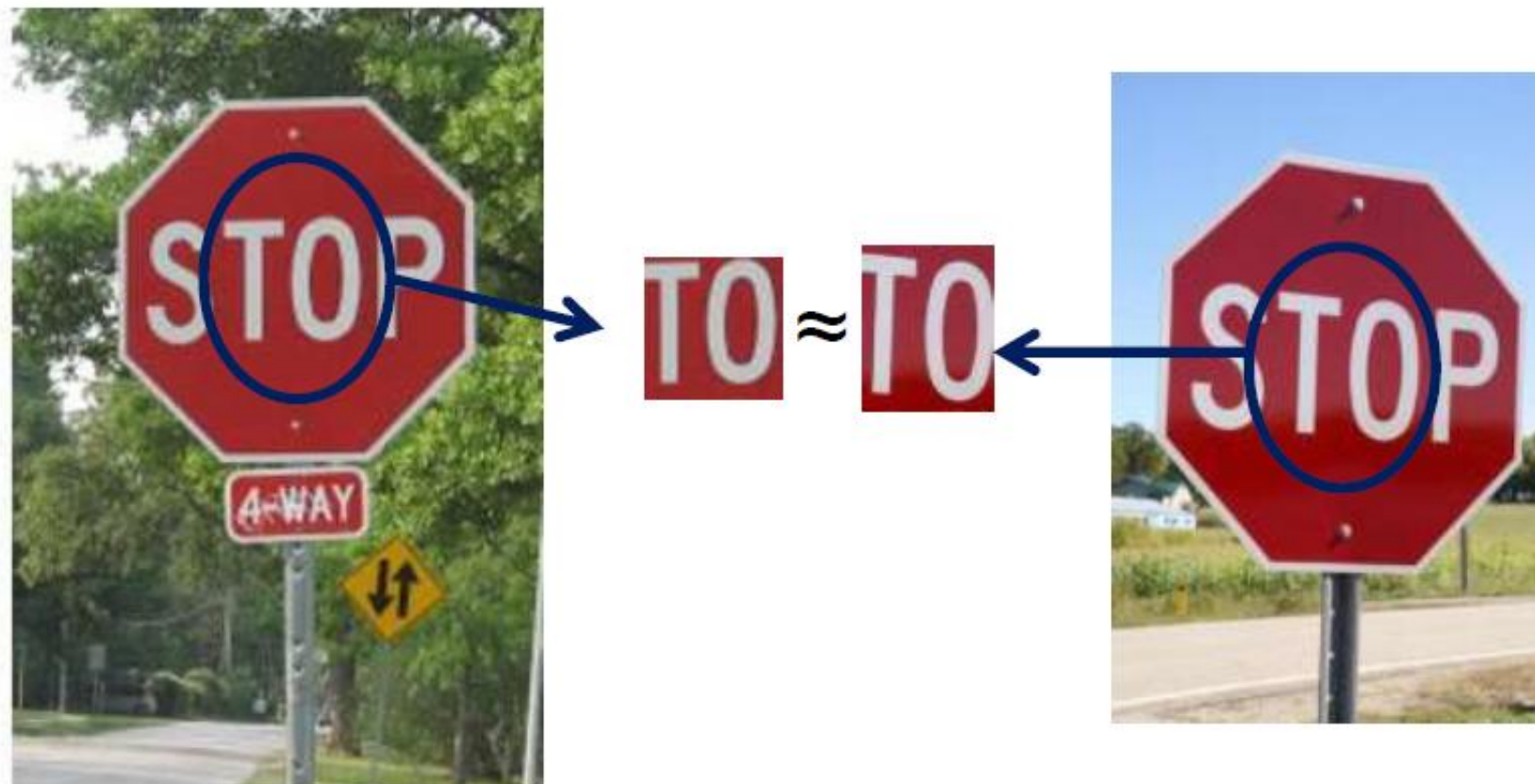


"Image created with a mobile phone" by Olaf Simons - Own work.



Correspondence and alignment

- Correspondence
 - Matching points, patches, edges, or regions across images



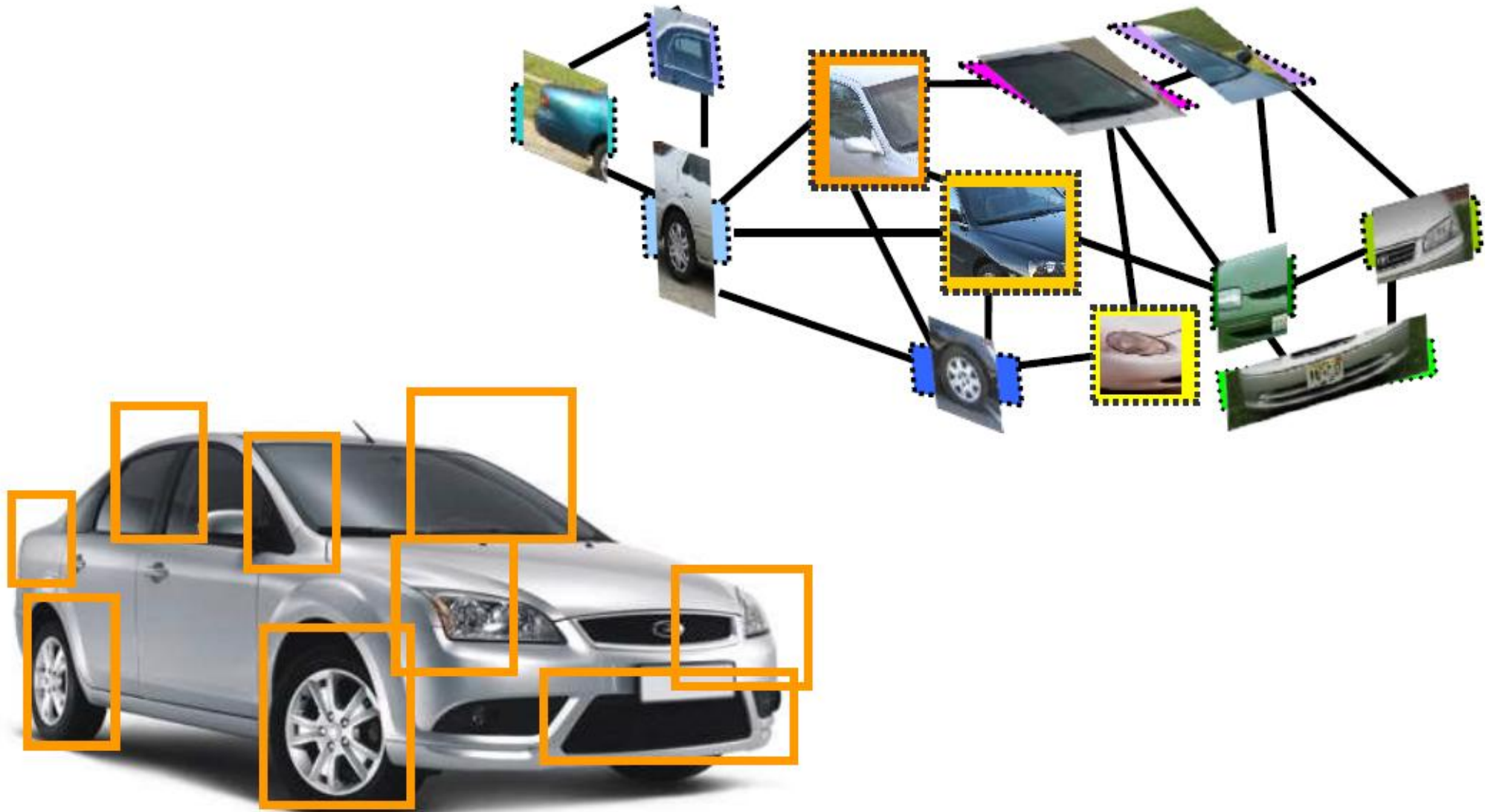
Correspondence and alignment



T

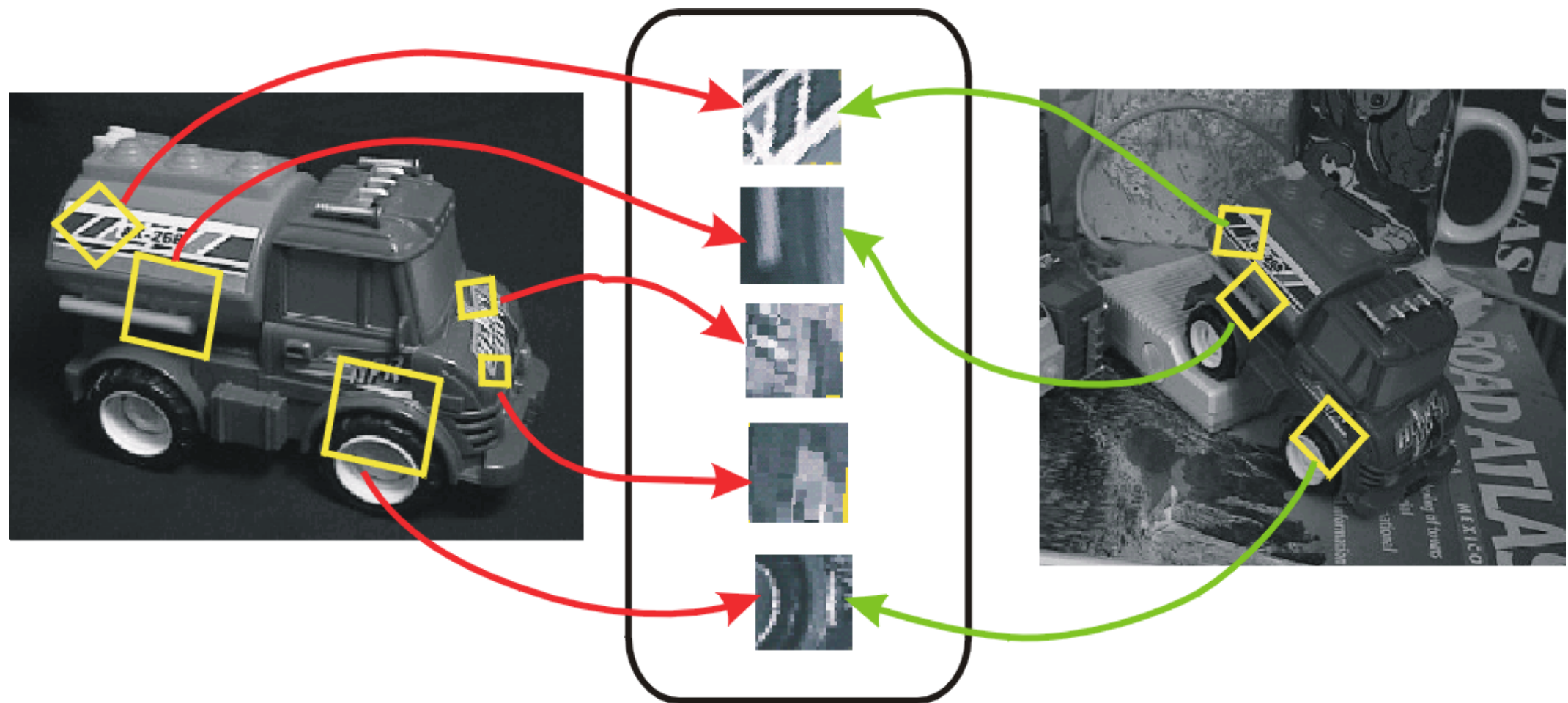


Fitting a 3D object model

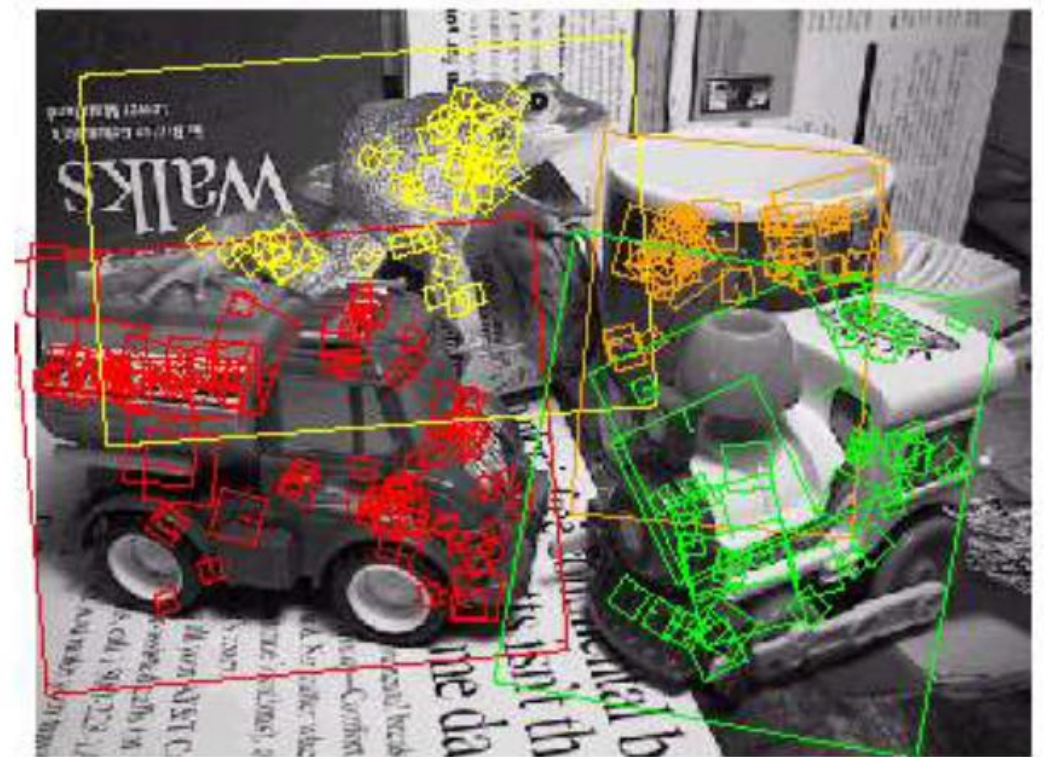
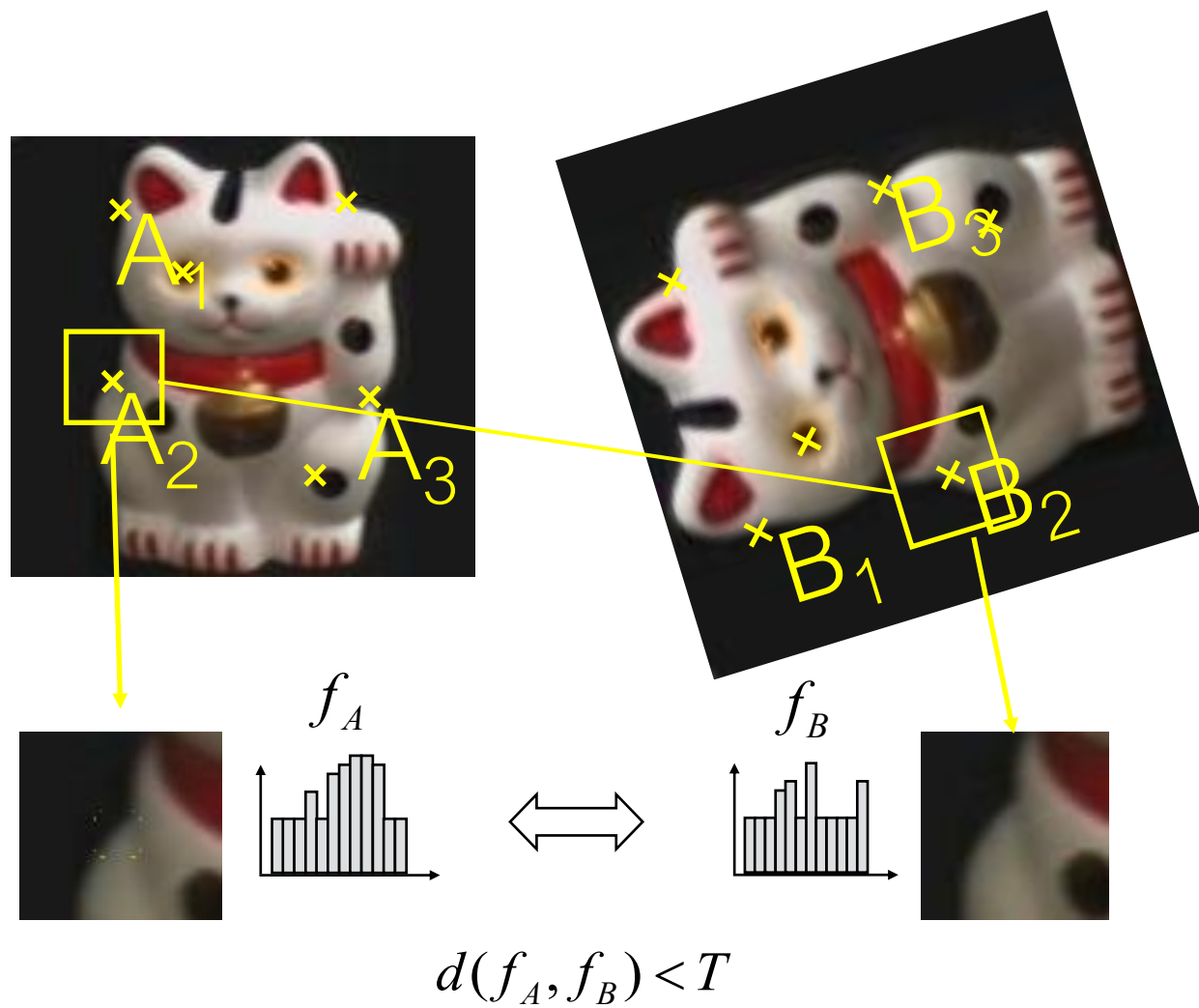


Invariant Local Features

- Image content is transformed into local feature coordinates that are invariant to translation, rotation, scale, and other imaging parameters

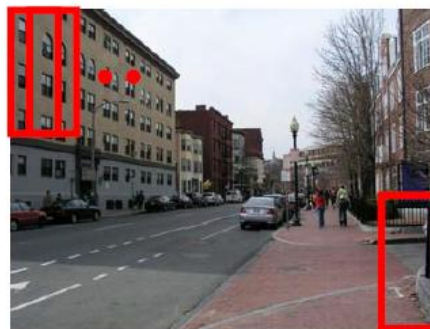


Keypoint matching



Object detection with sliding windows

- Find target objects for all possible image regions



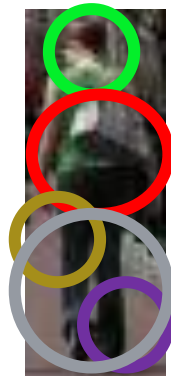
Statistical template

- Object model = sum of scores of features at fixed positions



$$+3 +2 -2 -1 -2.5 = -0.5 \stackrel{?}{>} 7.5$$

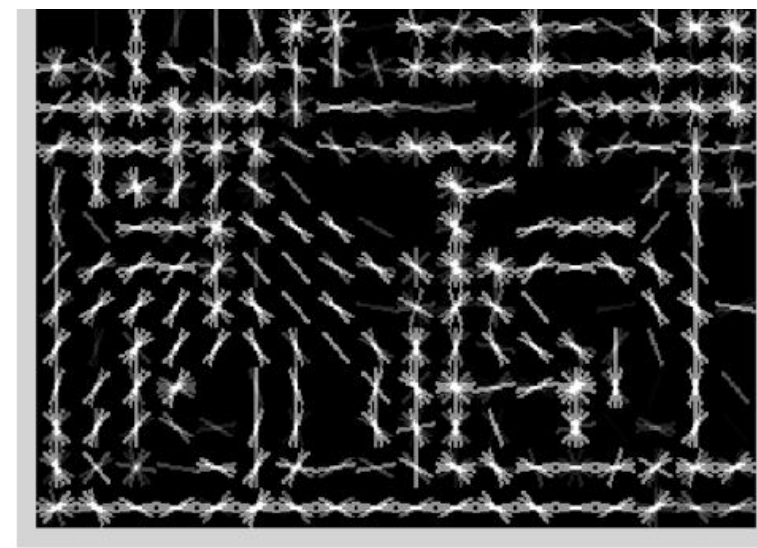
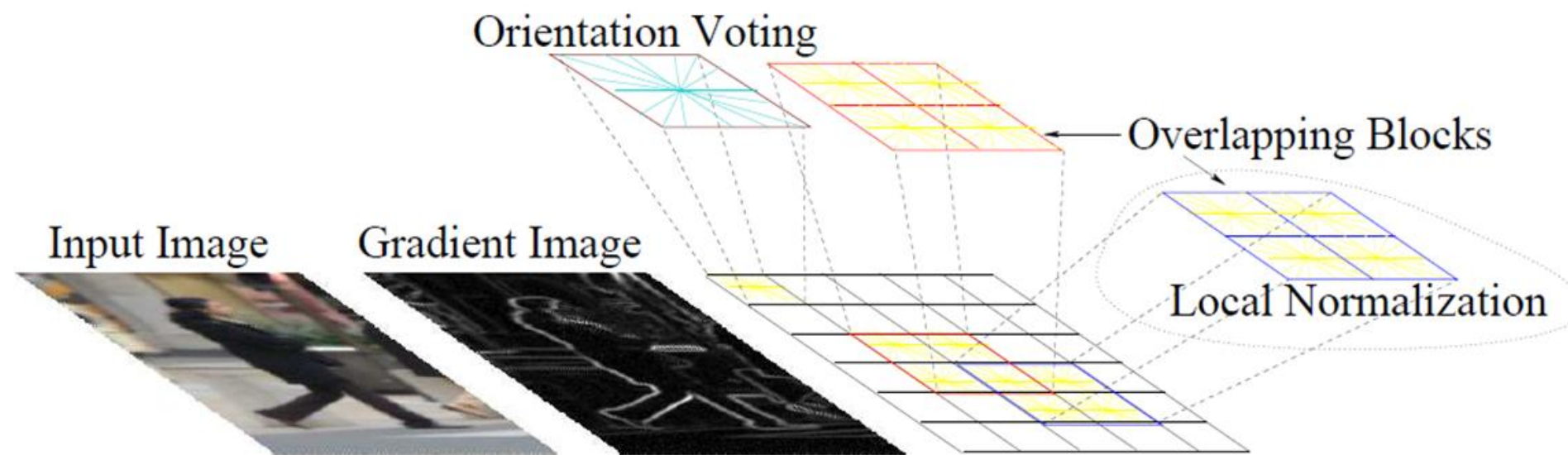
Non-object



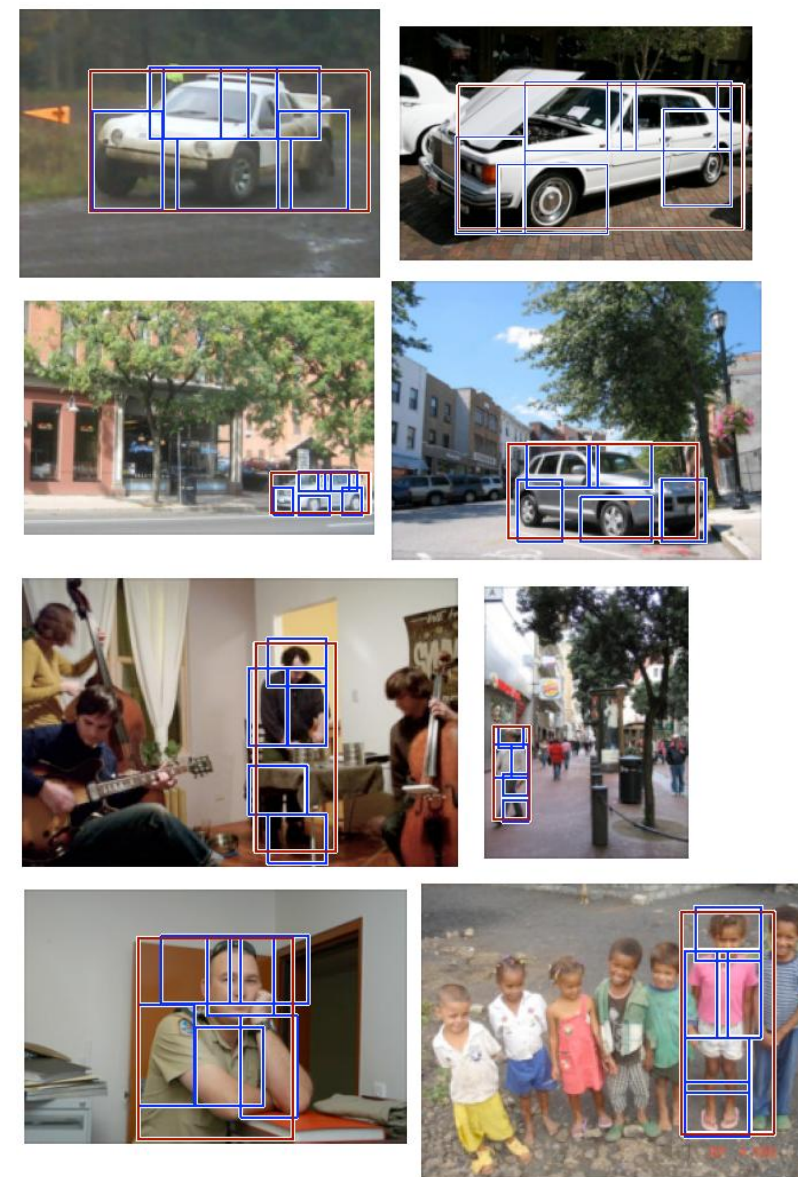
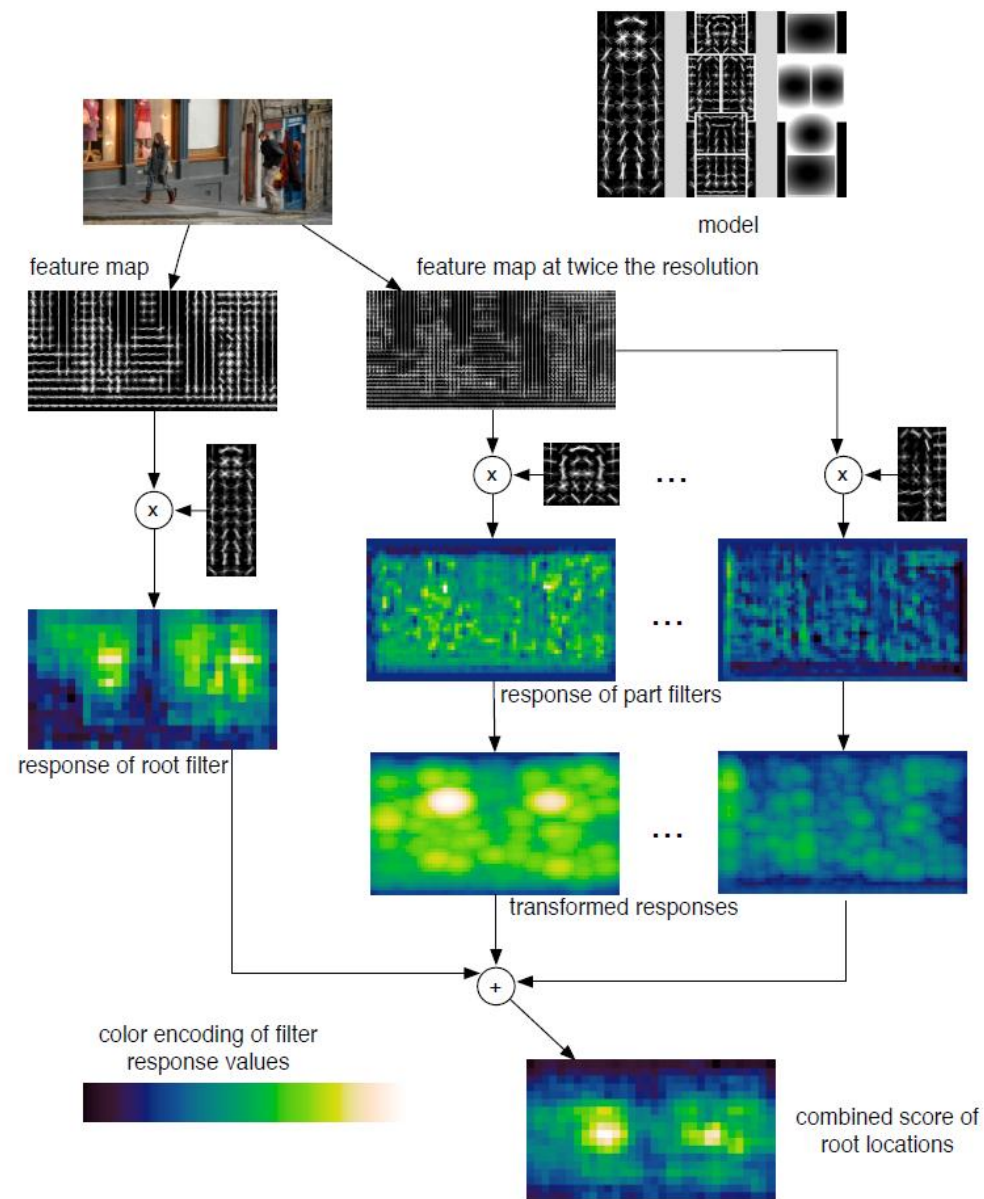
$$+4 +1 +0.5 +3 +0.5 = 10.5 \stackrel{?}{>} 7.5$$

Object

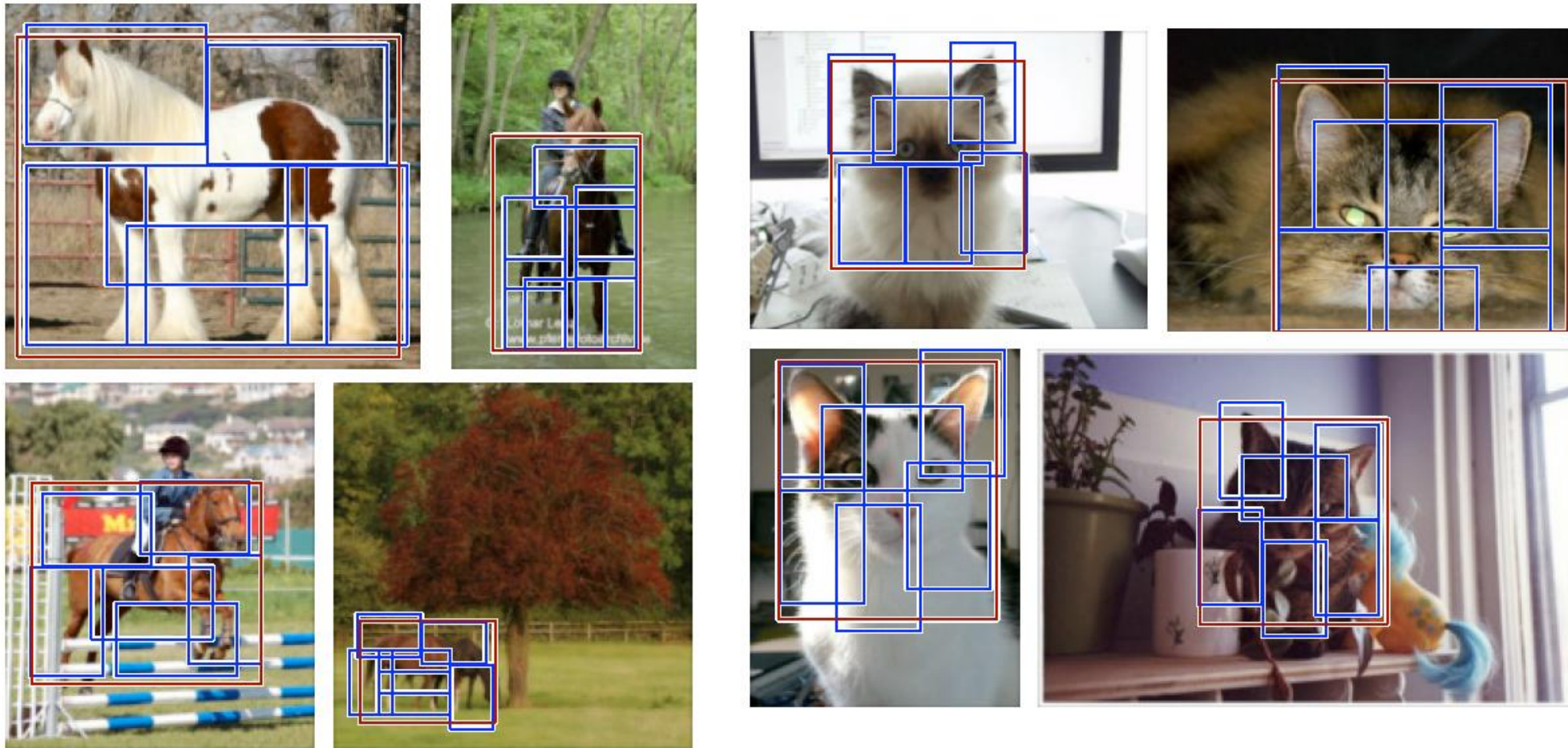
Histogram of Oriented Gradient (HOG) features



Part based models for deformable objects



Object detection results

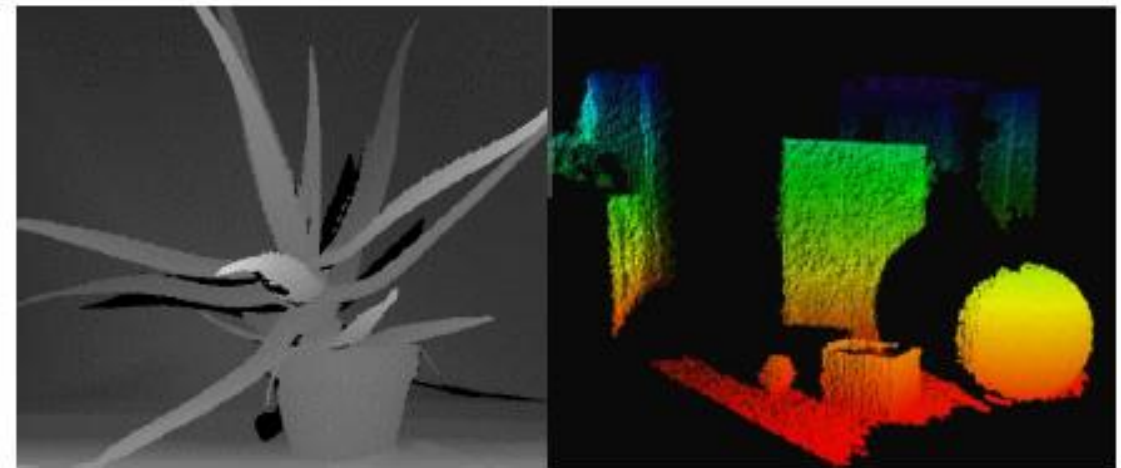


3D sensing

- Estimating depth or distance from a sensor to the



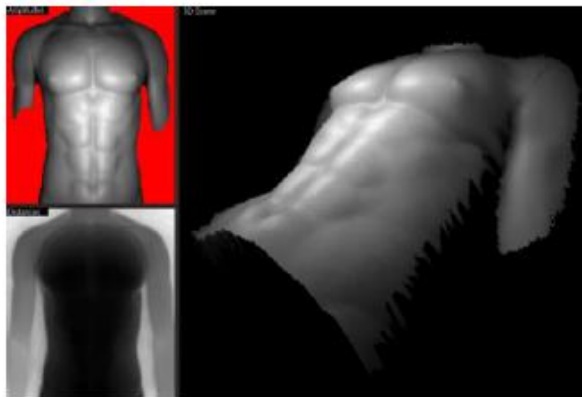
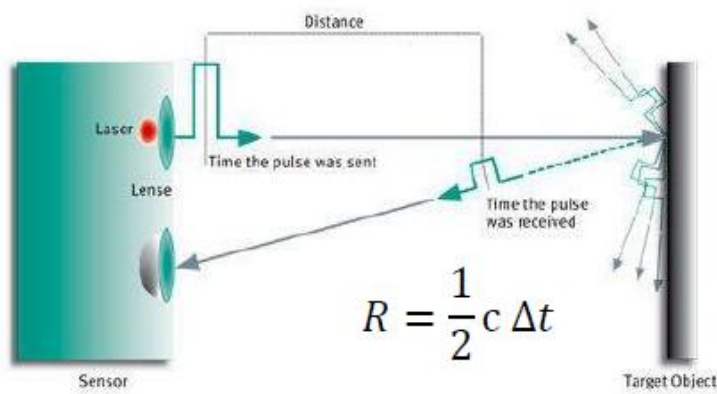
3d sensing with laser scanner



3D sensing results using stereo vision

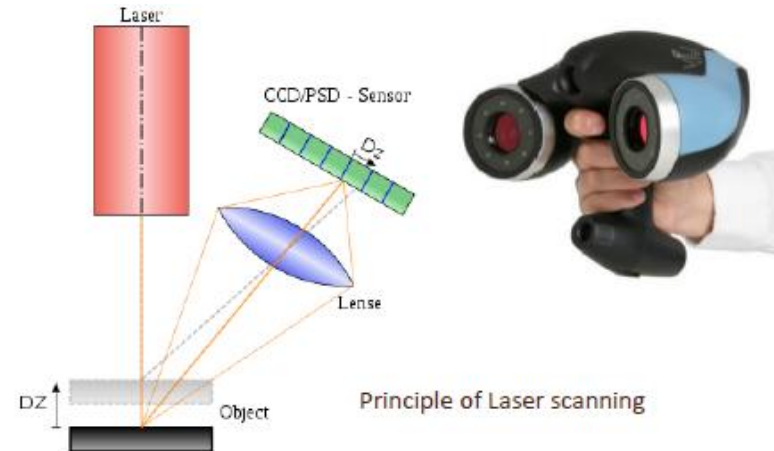
3D Sensing Methods in Metrology

ToF (Time-Of-Flight)



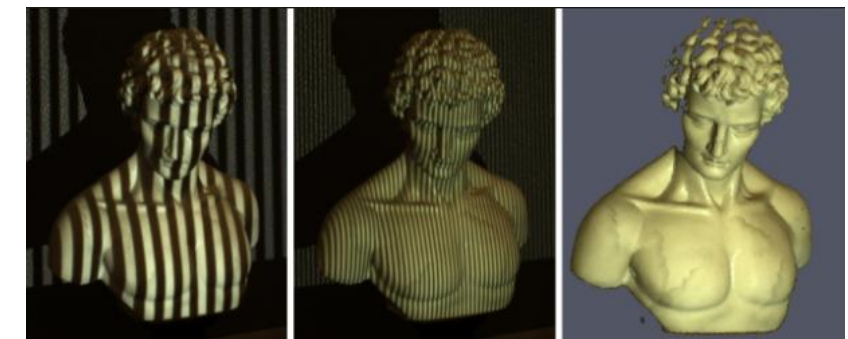
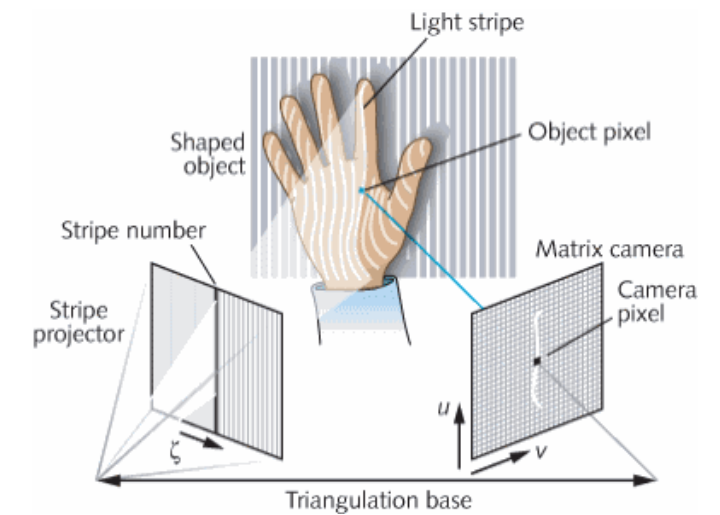
Principle of ToF sensors and acquired 3D data

Laser Scanner



Digital Michelangelo Project
<http://graphics.stanford.edu/projects/mich>

Structured light

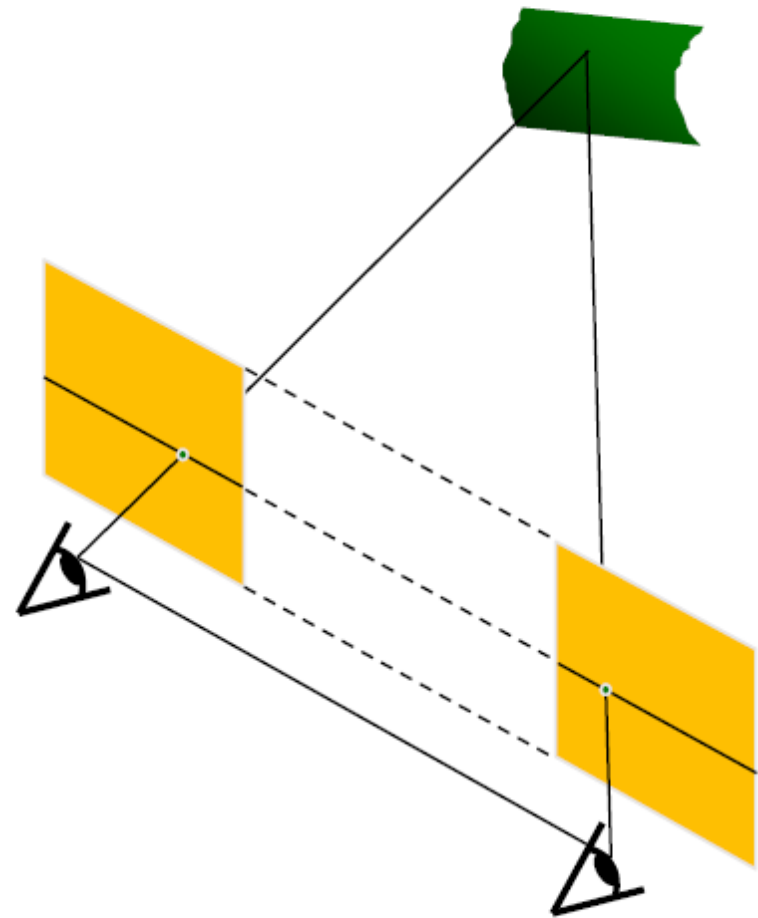


Stereo cameras



3D computation in stereo vision

Given projections of a 3D point in two or more images (with known camera matrices), finding the coordinates of the point



Stereo vision

- Triangulate on two images of the same point to recover depth
 - Feature matching across views
 - Calibrated cameras



Stereo vision

- Reconstruction / triangulation

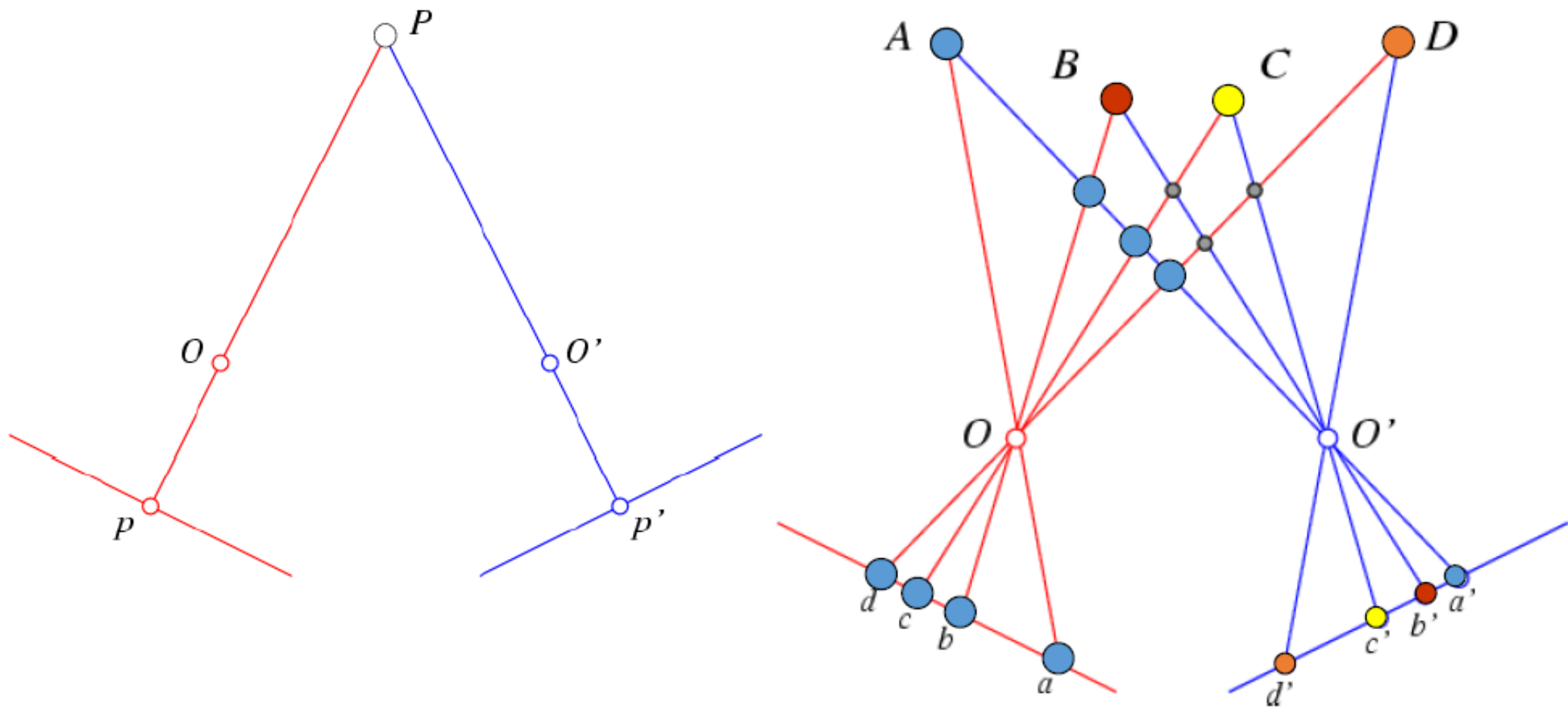
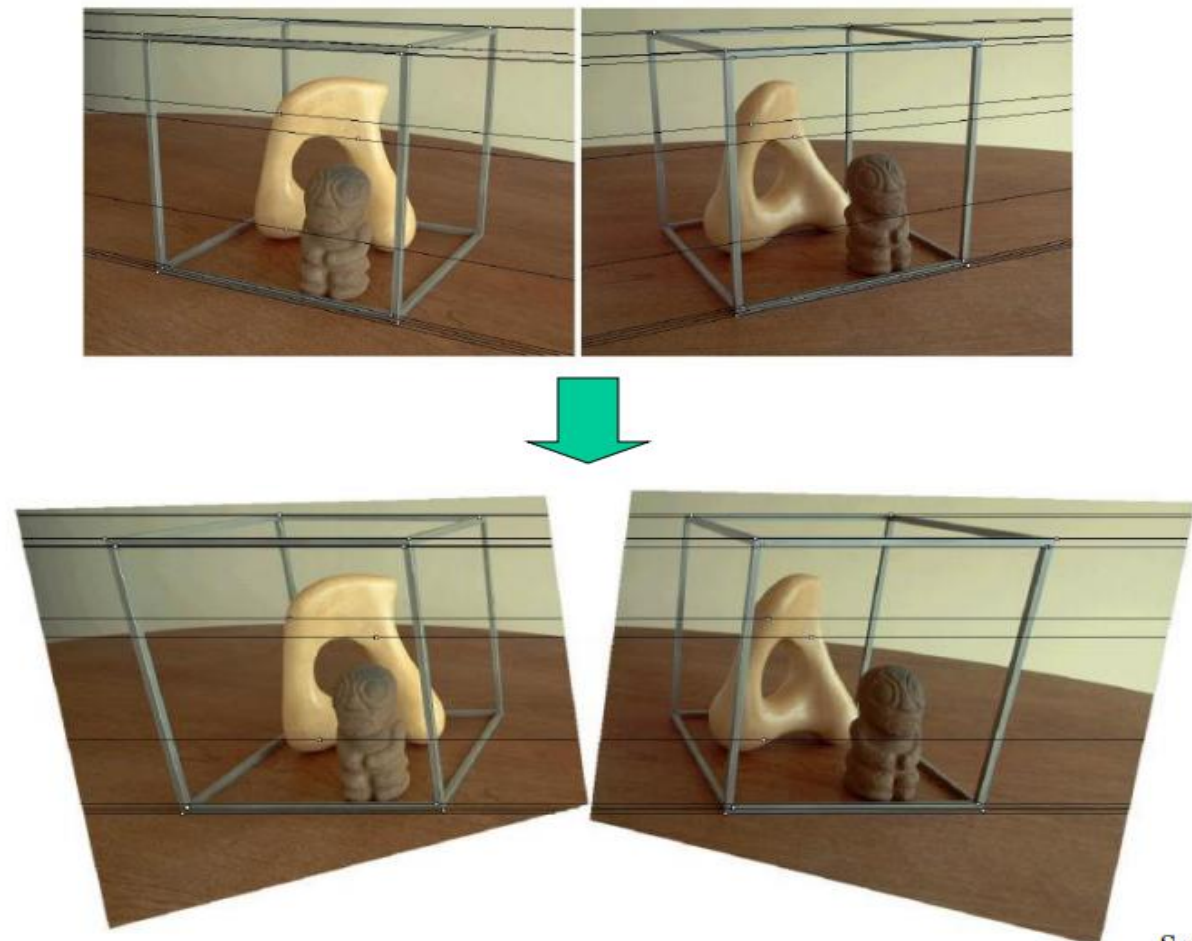
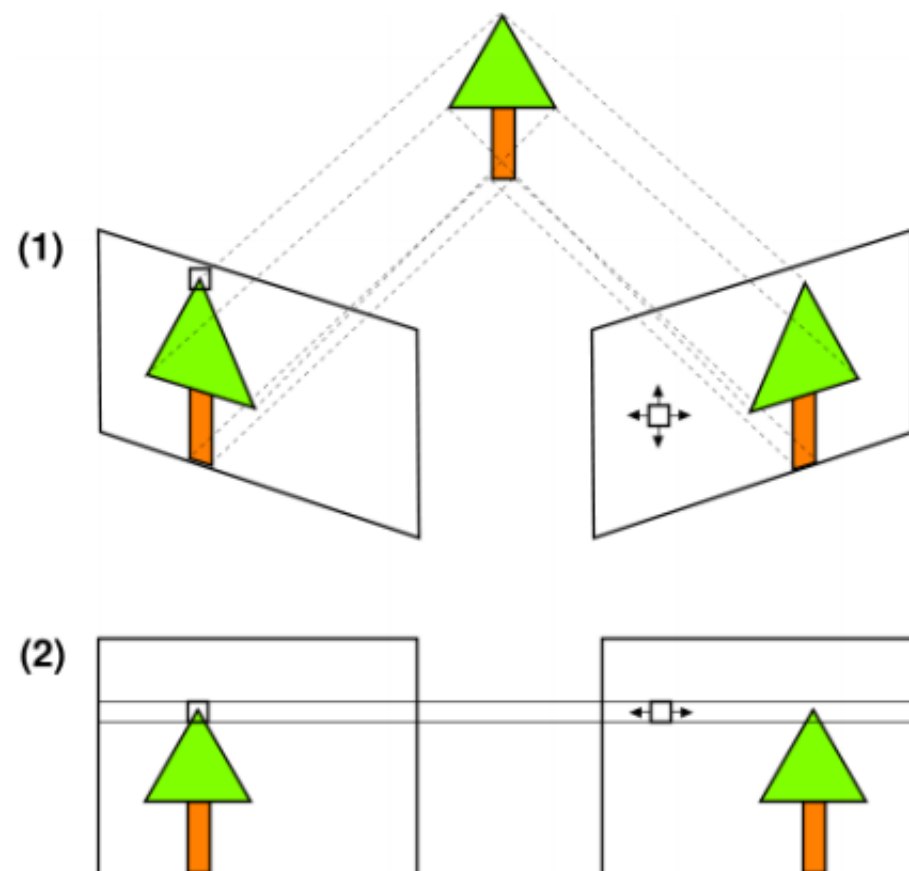
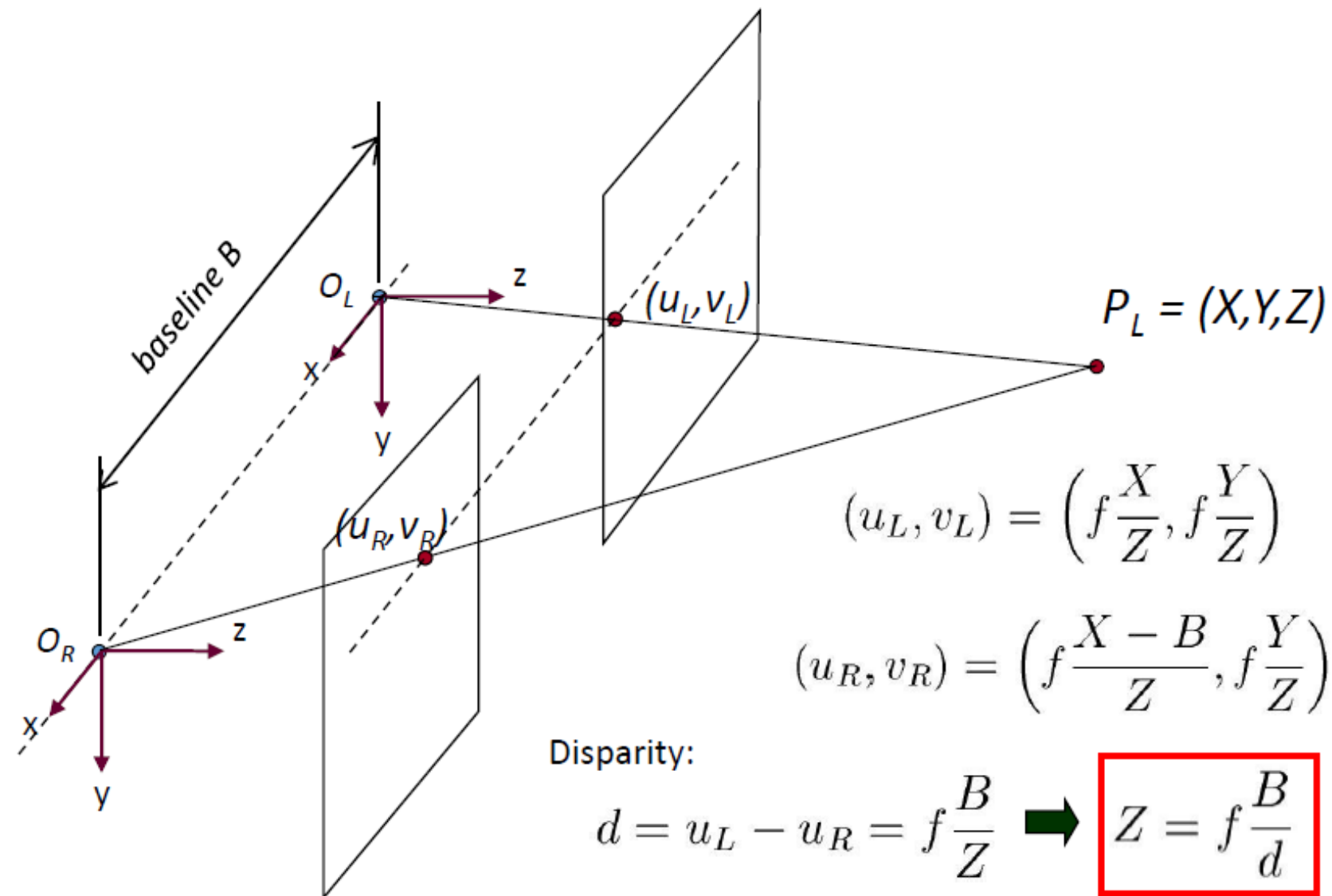


Image rectification

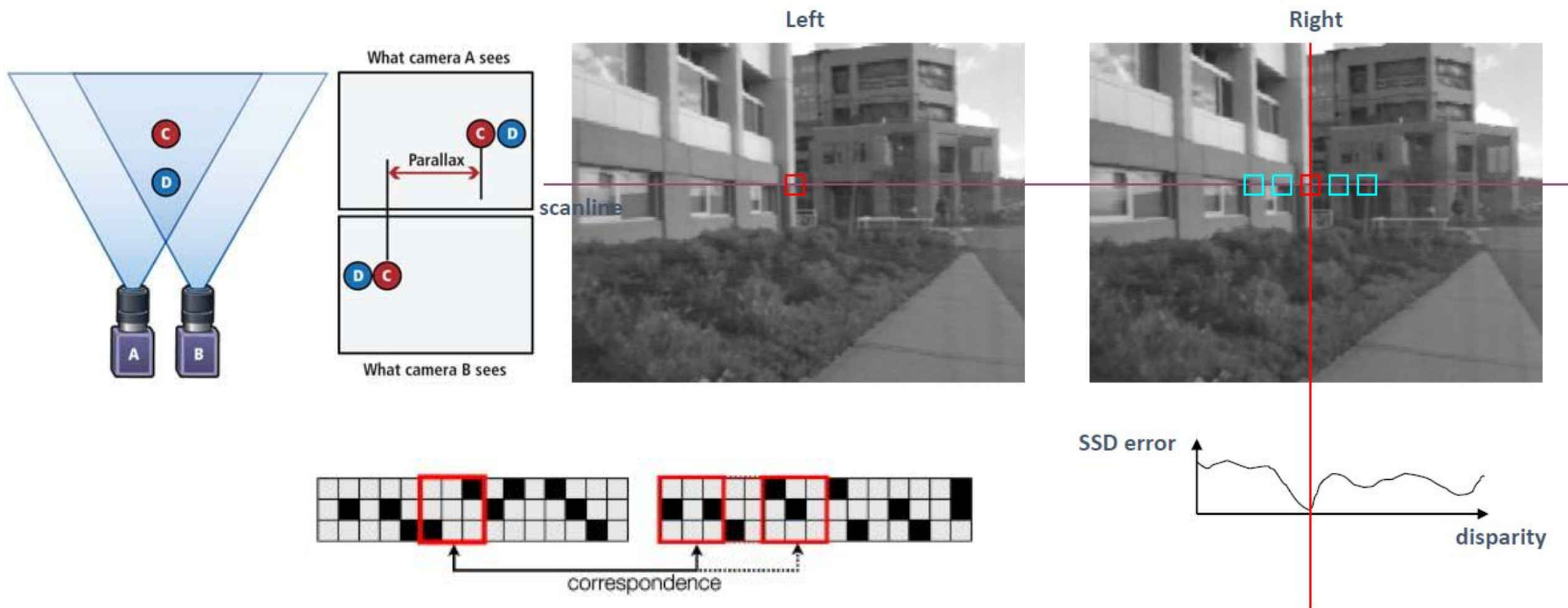
- If the two cameras are aligned to be coplanar, the search is simplified to one dimension
 - A horizontal line parallel to the baseline between the cameras



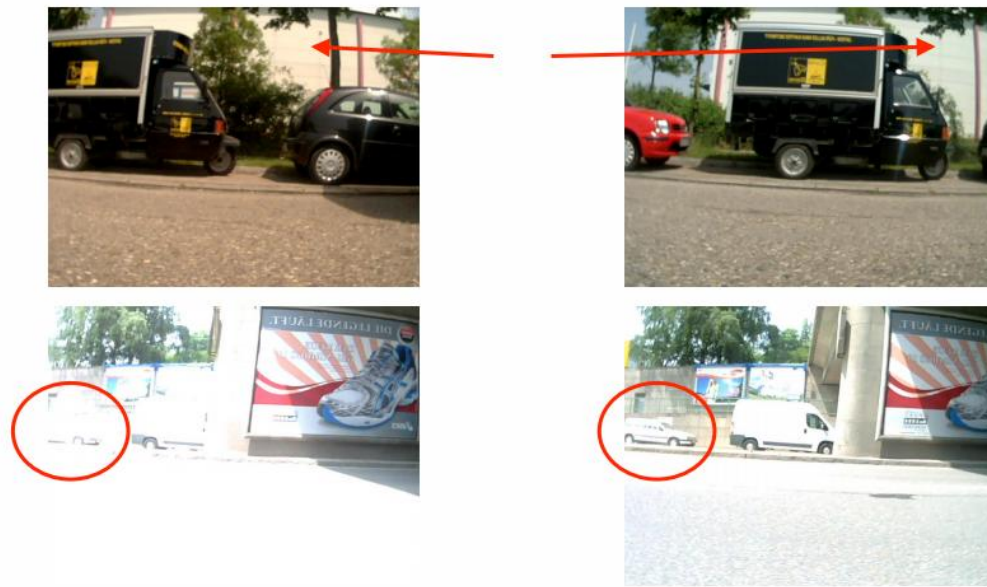
Depth estimation with rectified images



Similarity measure



Challenges for stereo matching



Photometric Variations



Specularities

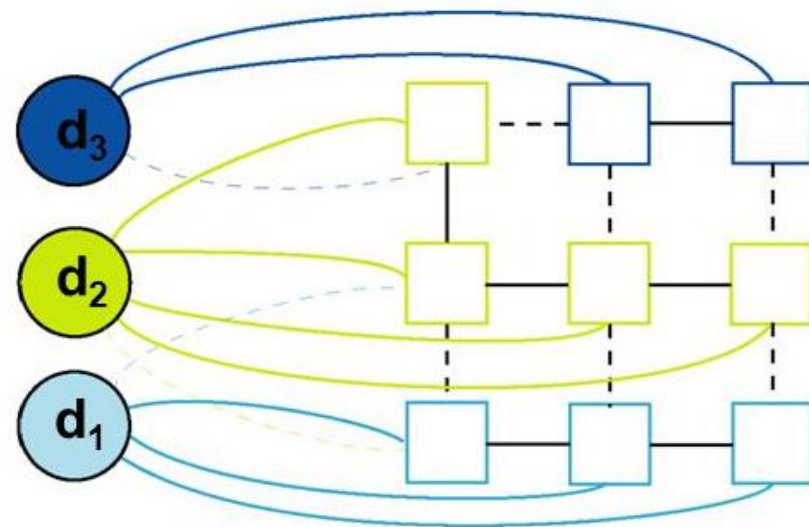


Occlusions

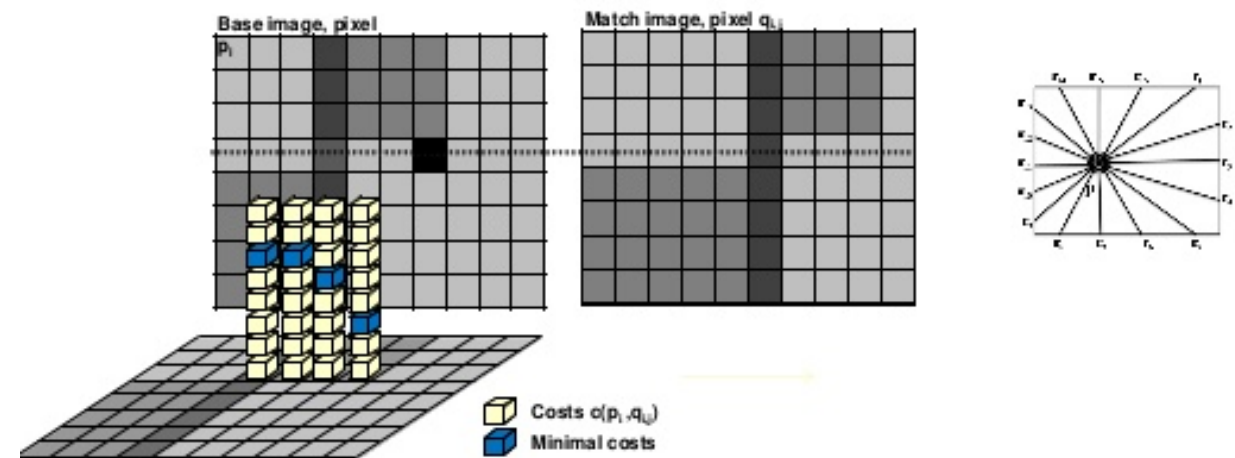


Transparency

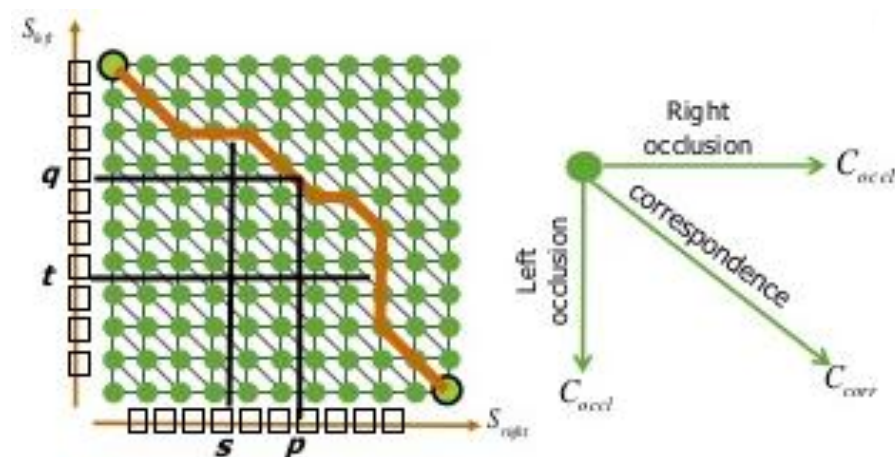
Stereo matching with global optimization



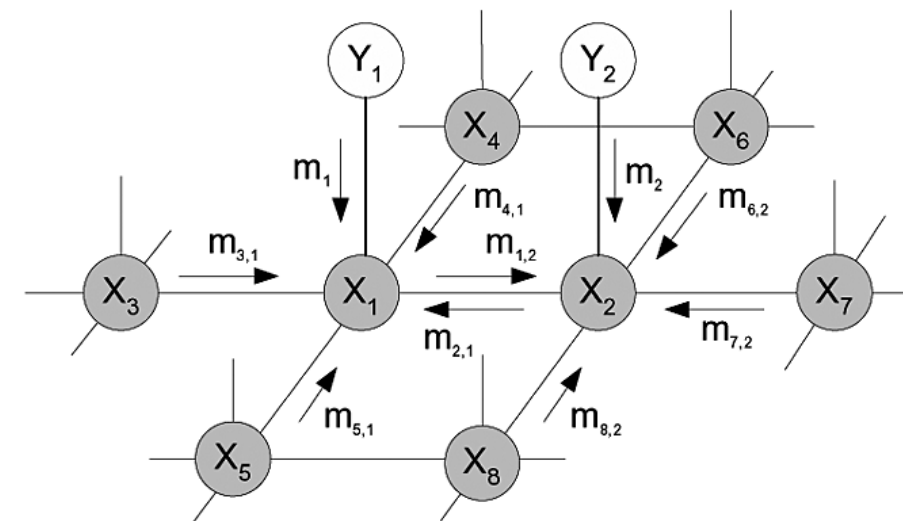
Graph cuts



Semi-Global Matching



Dynamic programming



Belief Propagation

Stereo matching results

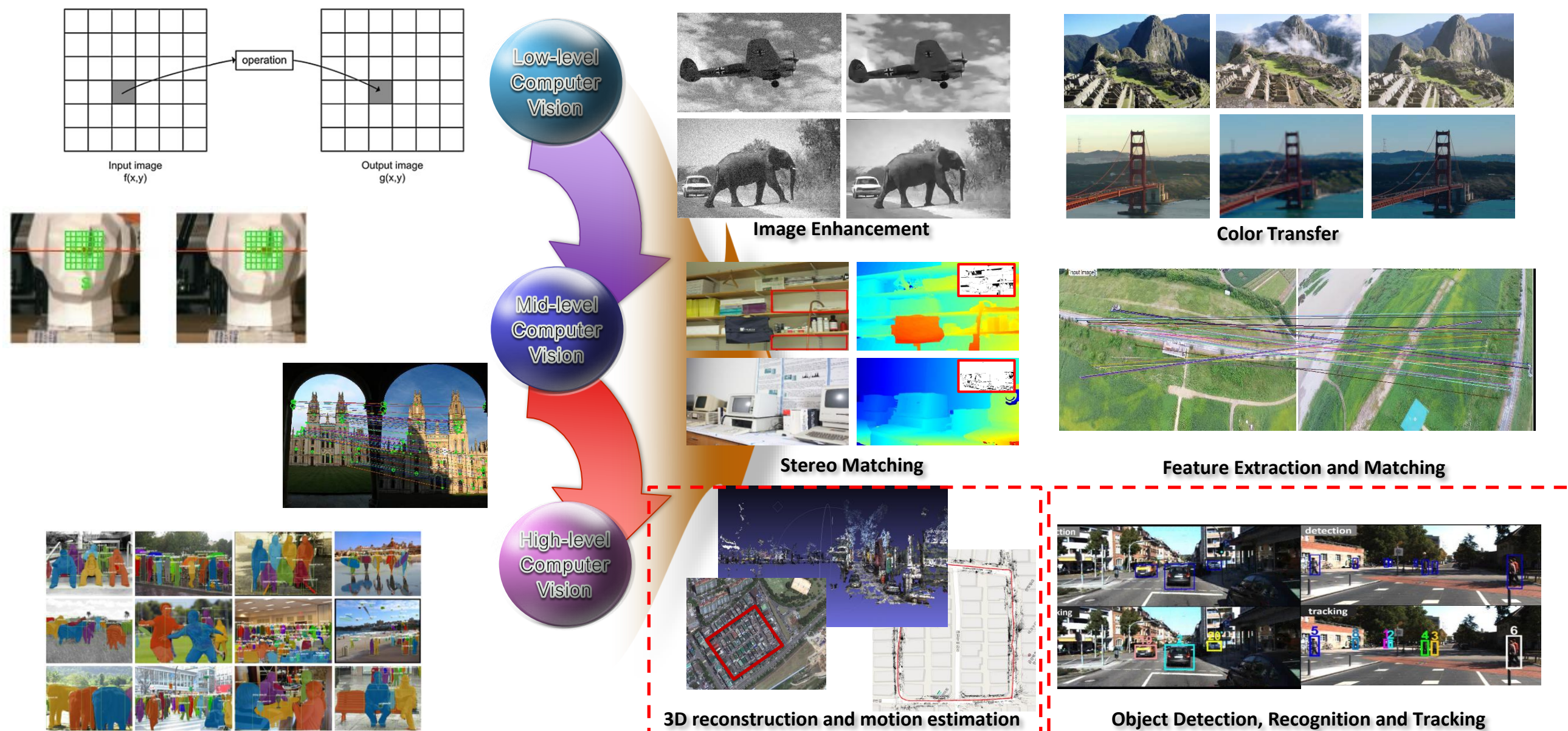
Input



Depth



Computer Vision Categories

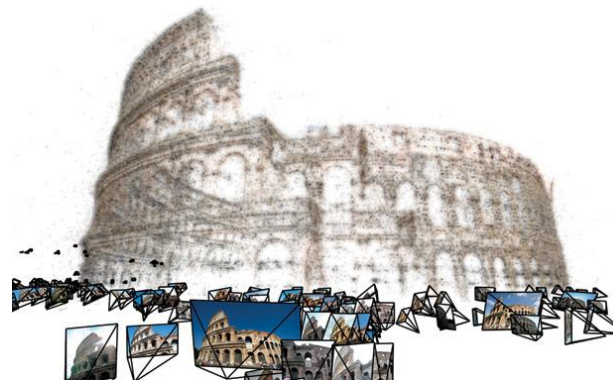


Main Goals of Computer Vision



What (who)?

Object (Face) Recognition



Where?

3D reconstruction
Pose estimation



A man is pouring oil into a pot.



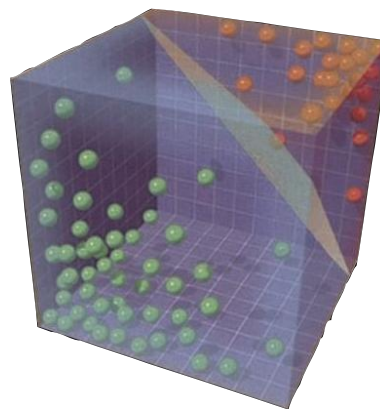
*The person opened the drawer.
The person took out a pot.
The person went to the sink.
The person washed the pot.
The person turned on the stove.*

How?

Video Captioning

typical perception pipeline

representation



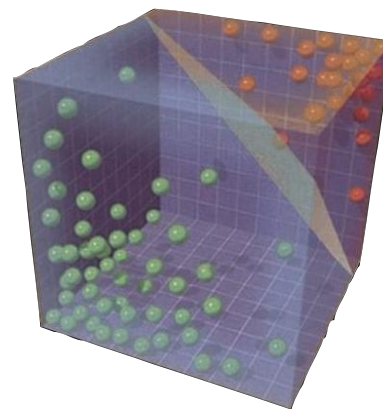
'fancy math'



output

typical perception pipeline

representation



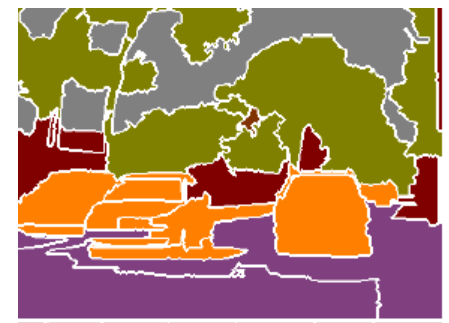
'fancy math'



output

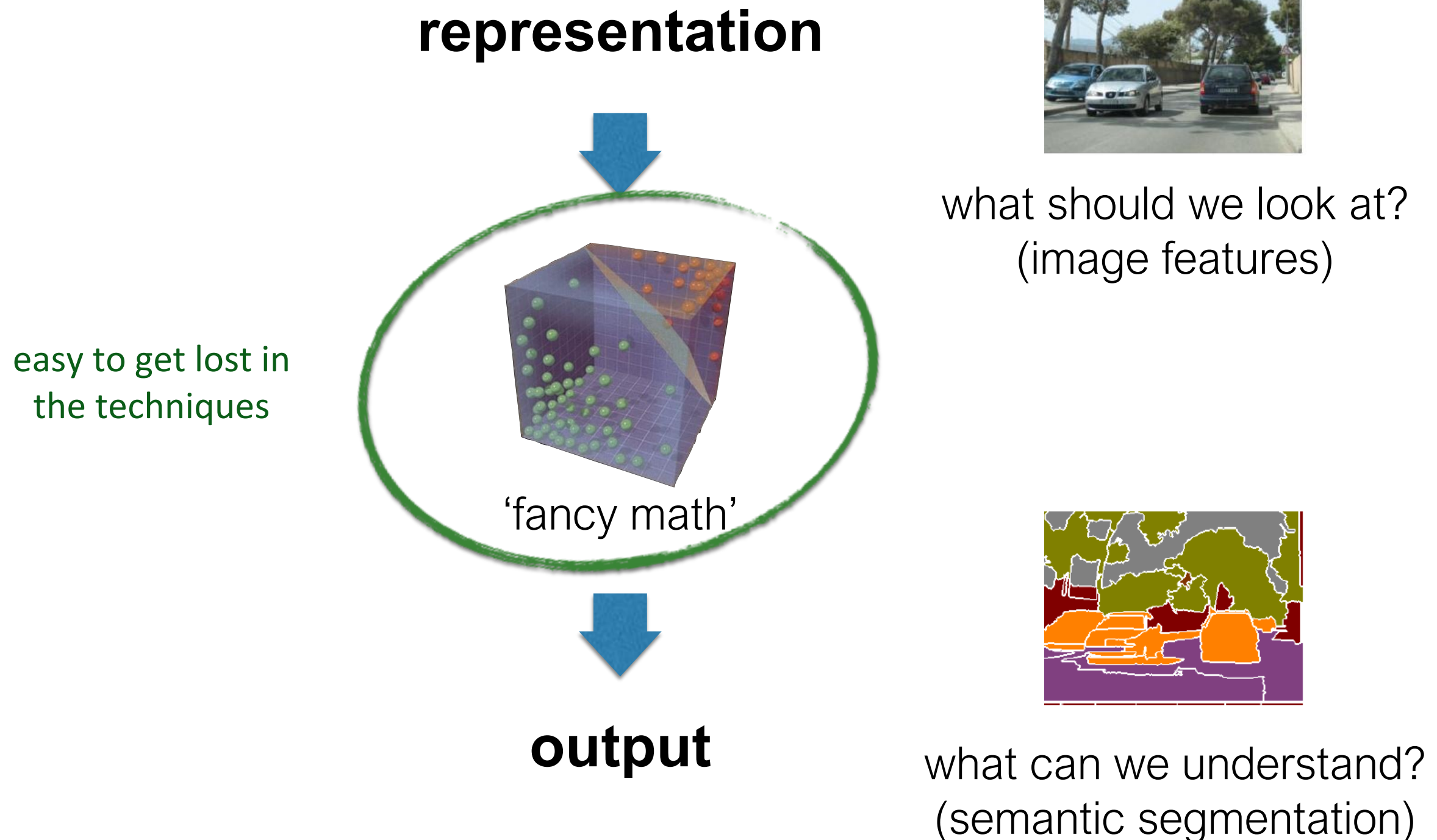


what should we look at?
(image features)

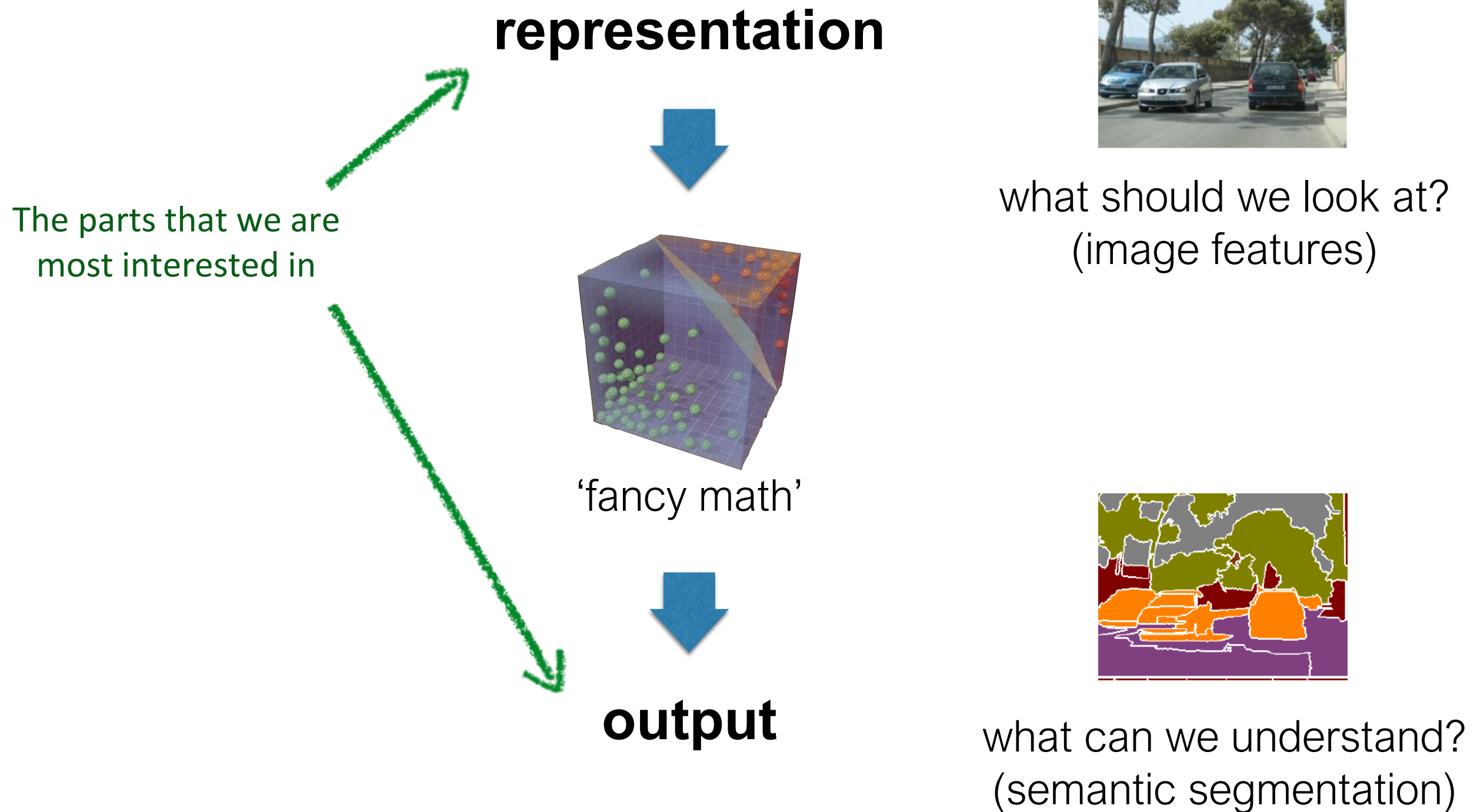


what can we understand?
(semantic segmentation)

typical perception pipeline



typical perception pipeline



Important note:

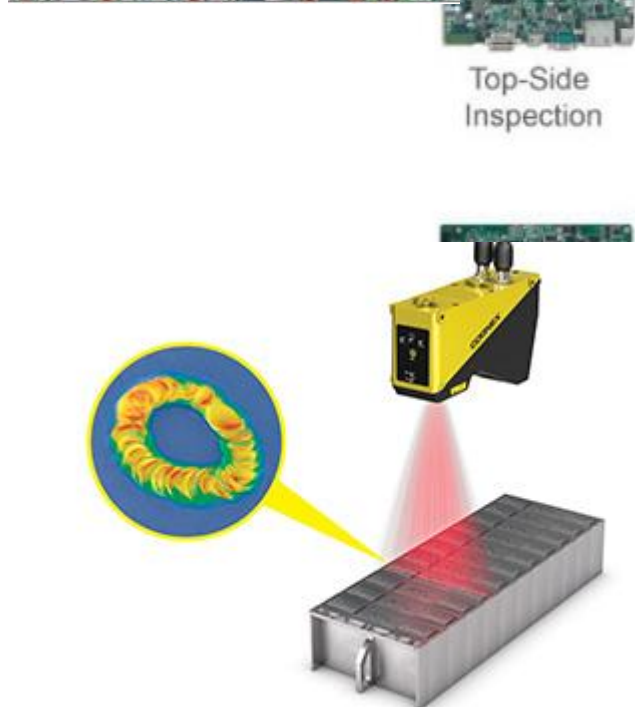
In general, computer vision does not work

Important note:

In general, computer vision does not work
(except in certain situations/conditions)

Applications of computer vision

Automated visual inspection



The diagram illustrates the MIO-2270 2.5" MIO-Ultra Single Board Computer. It features a central green PCB populated with various components, including a CPU, memory modules, and I/O connectors. The board is shown in a perspective view, highlighting its compact size. To the left, two blue industrial high-speed cameras are connected to the board via red lines labeled "USB3.0". To the right, a green line labeled "GPIO" and a blue line labeled "Gigabit Ethernet" indicate additional connectivity options. The board is shown in a perspective view, highlighting its compact size.

Industrial High-Speed Camera

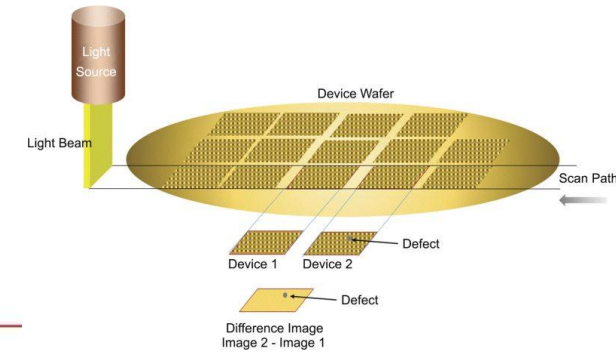
Industrial High-Speed Camera

USB3.0

GPIO

Gigabit Ethernet

MIO-2270
2.5" MIO-Ultra Single Board Computer

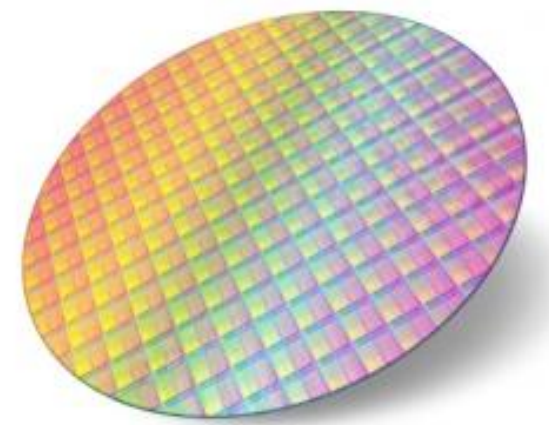


OPTOFIDELITY DISPLAY INSPECTION



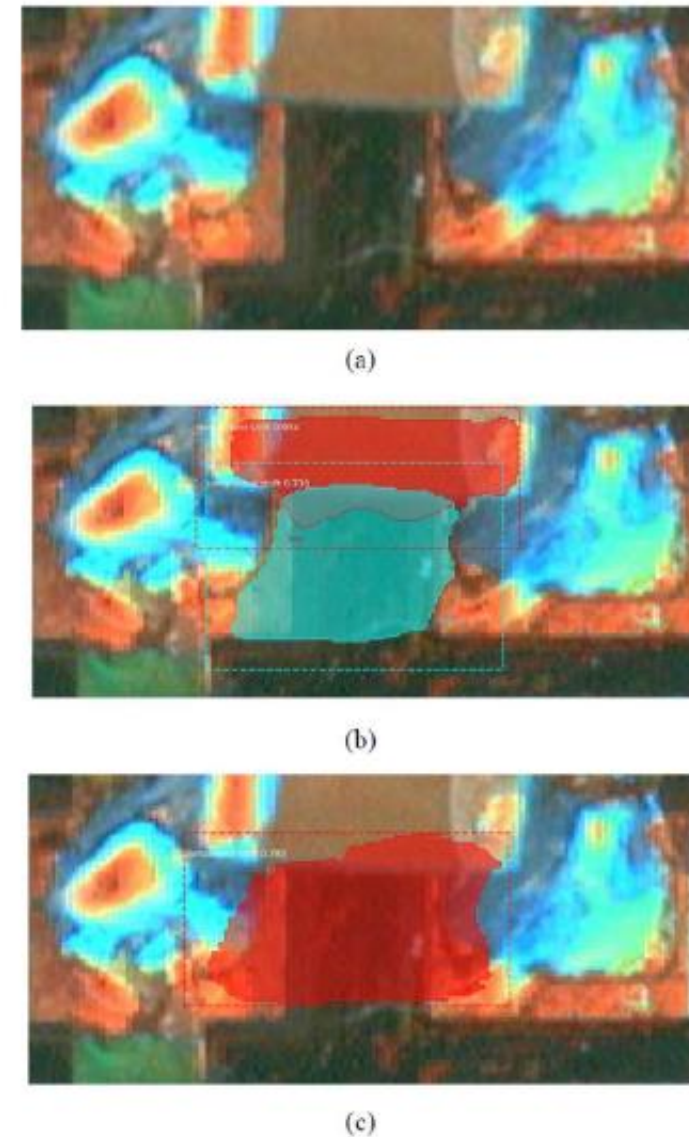
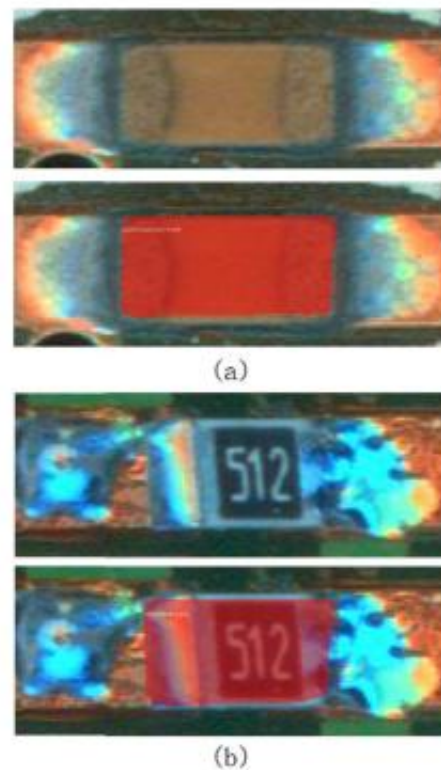
Machine vision

Automated visual inspection



Recent deep-learning based inspection system

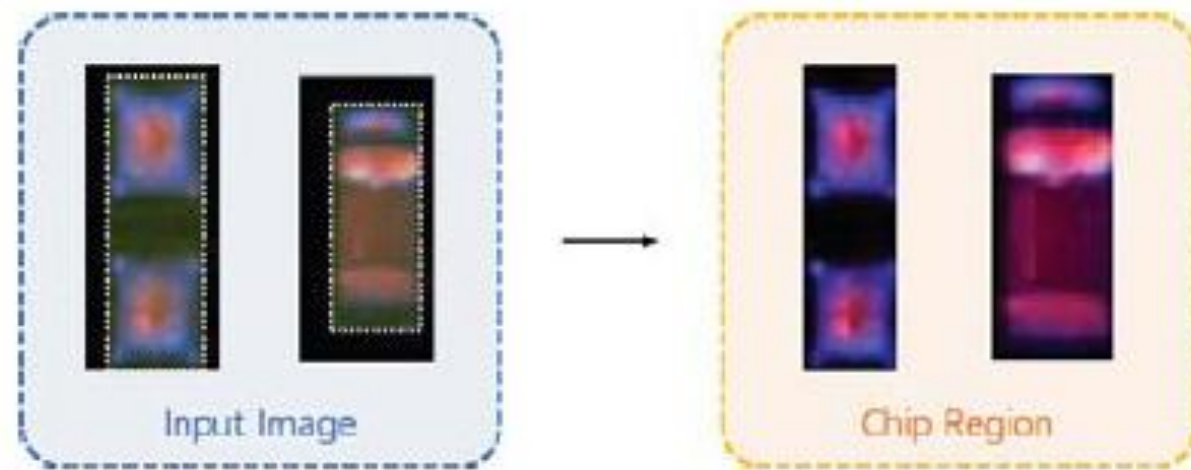
- Mask R-CNN based Solder Joint Recognition



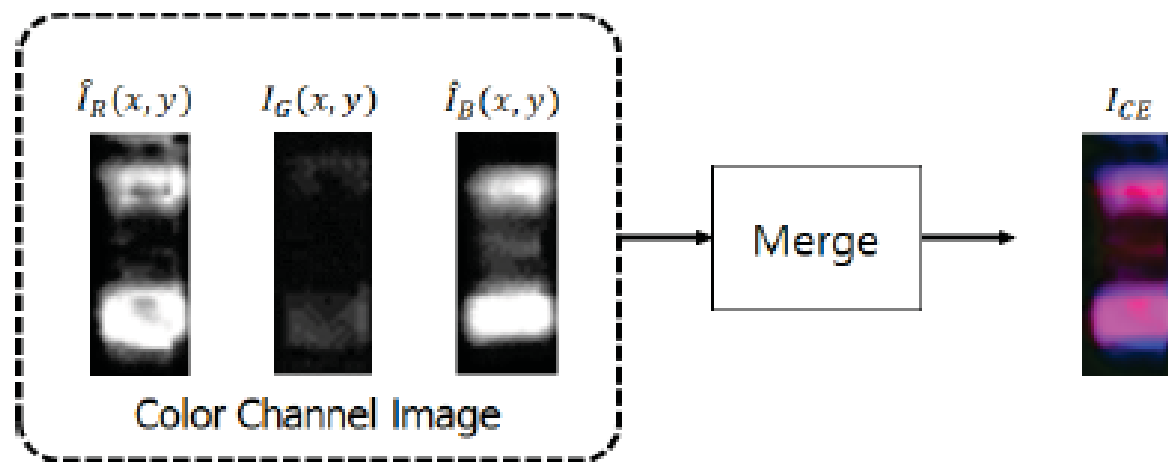
Method	Accuracy	mAP _{bbbox}
Eginsoder [6]	96.43%	---
Support vector machine [12]	96.43%	---
Ensemble learning [13]	89.29%	---
Our proposed method	100%	97.4%

Recent deep-learning based inspection system

- CNN based AOI defect classification

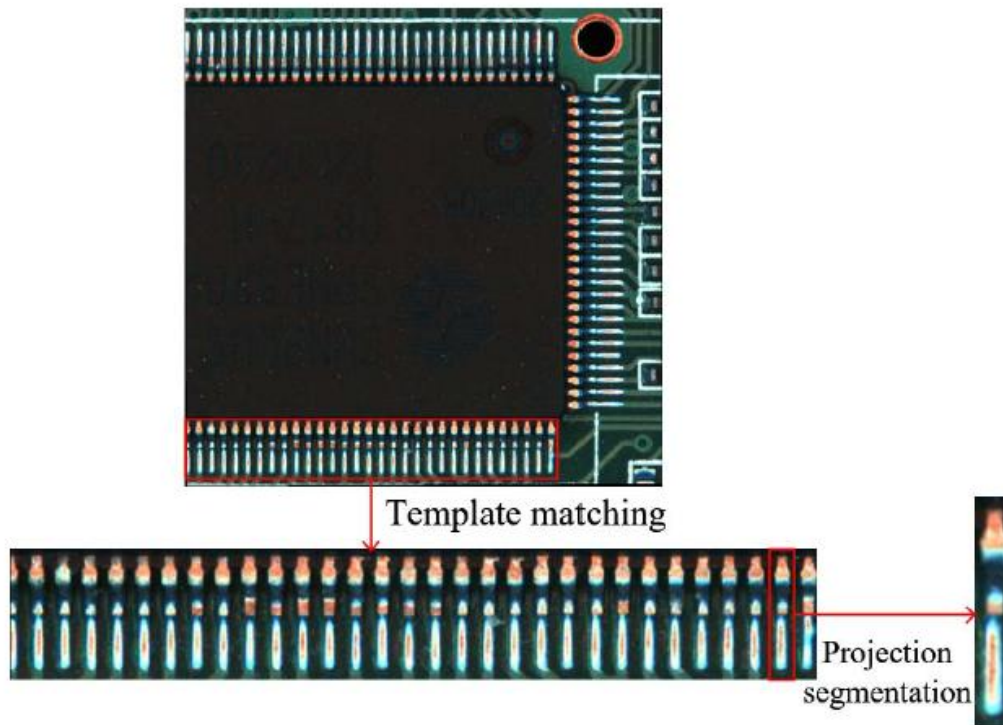
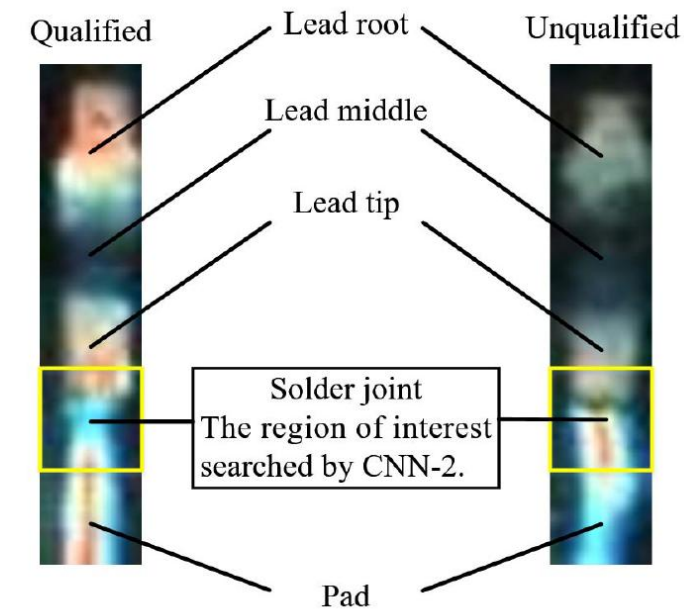
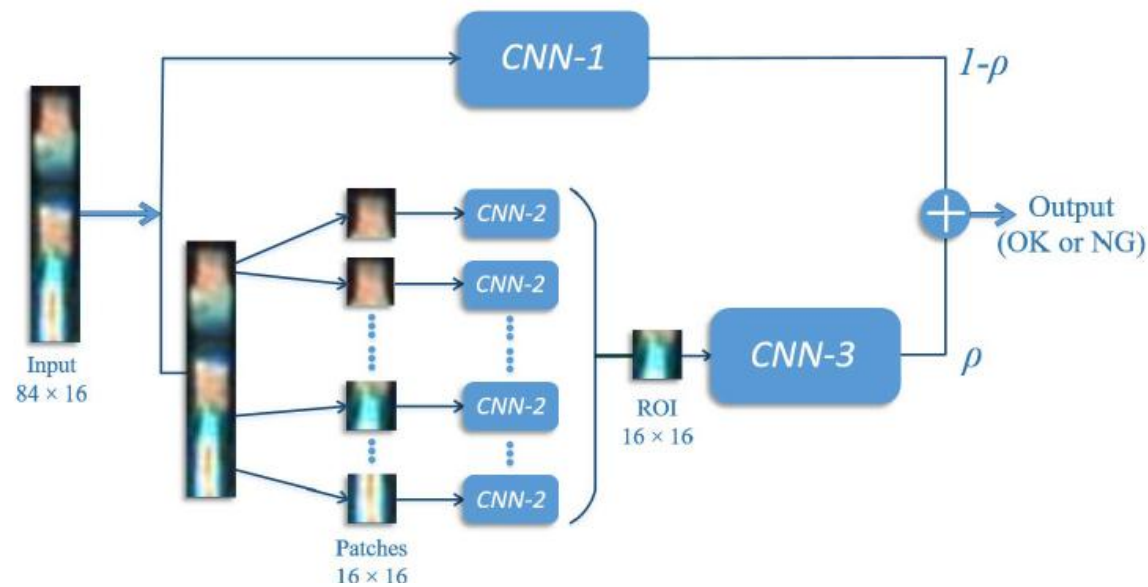


Defect Type	Method		
	<i>CNN</i>	<i>CNN with CRE</i>	<i>CNN with CRE and CE (Proposed)</i>
Exceed	94.4	94.4	94.4
Manhattan	100.0	100.0	97.1
Missing	86.7	96.7	92.9
Normal C	95.5	99.3	99.5
Normal R	81.3	87.4	97.6
No-Solder	99.1	91.5	94.2
Pseudo	85.5	92.1	79.0
Rotation	41.0	69.1	76.5
Shift	74.2	80.3	80.3
Tombstone	98.8	99.4	99.4
Average Accuracy	85.7	91.0	91.1



Defect Type	Method		
	<i>CNN</i>	<i>CNN with CRE</i>	<i>CNN with CRE and CE (Proposed)</i>
Precision	77.0	83.4	90.6
Recall	85.7	91.0	91.1
F1-Score	81.1	87.0	90.8

Recent deep-learning based inspection system



Method	Recall	Precision	F-measure
Bayes	0.6610	0.7402	0.6984
SVM	0.7210	0.7357	0.7283
G-MLP	0.7000	0.7600	0.7288
GMM	0.8520	0.9446	0.8959
ViBe	0.9660	0.8625	0.9113
ours	0.9980	1	0.9990

Amazon Go



Toshiba Tech IS-910T

2013

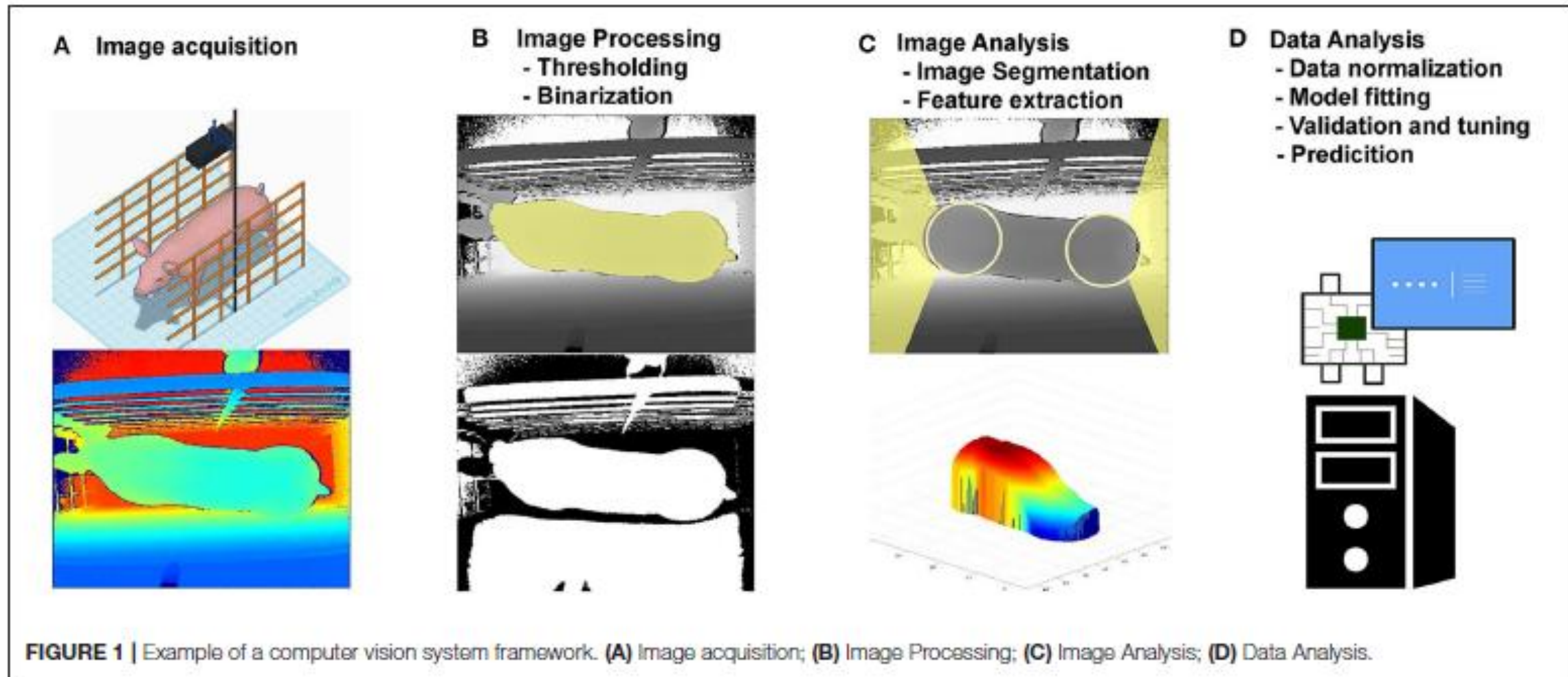


DataLogic LaneHawk LH4000

2012



Computer vision for Veterinary Science



A. F. A. Fernandes et al., Image Analysis and Computer Vision Applications in Animal Sciences: An Overview, frontiers in Veterinary Science, 2020

3D Reconstruction



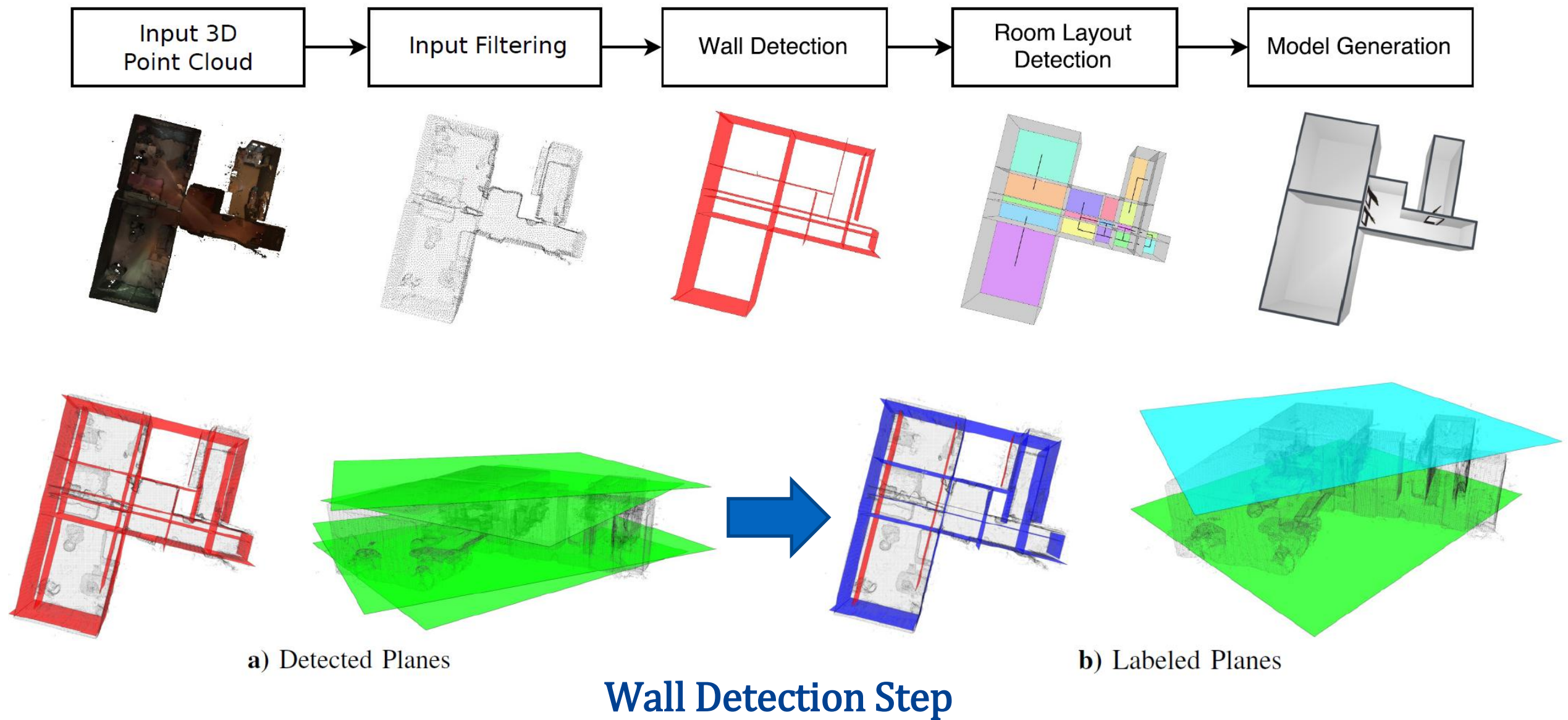
3D Reconstruction

VGGT: Visual Geometry Grounded Transformer
- Are we approaching a 3D foundation model?

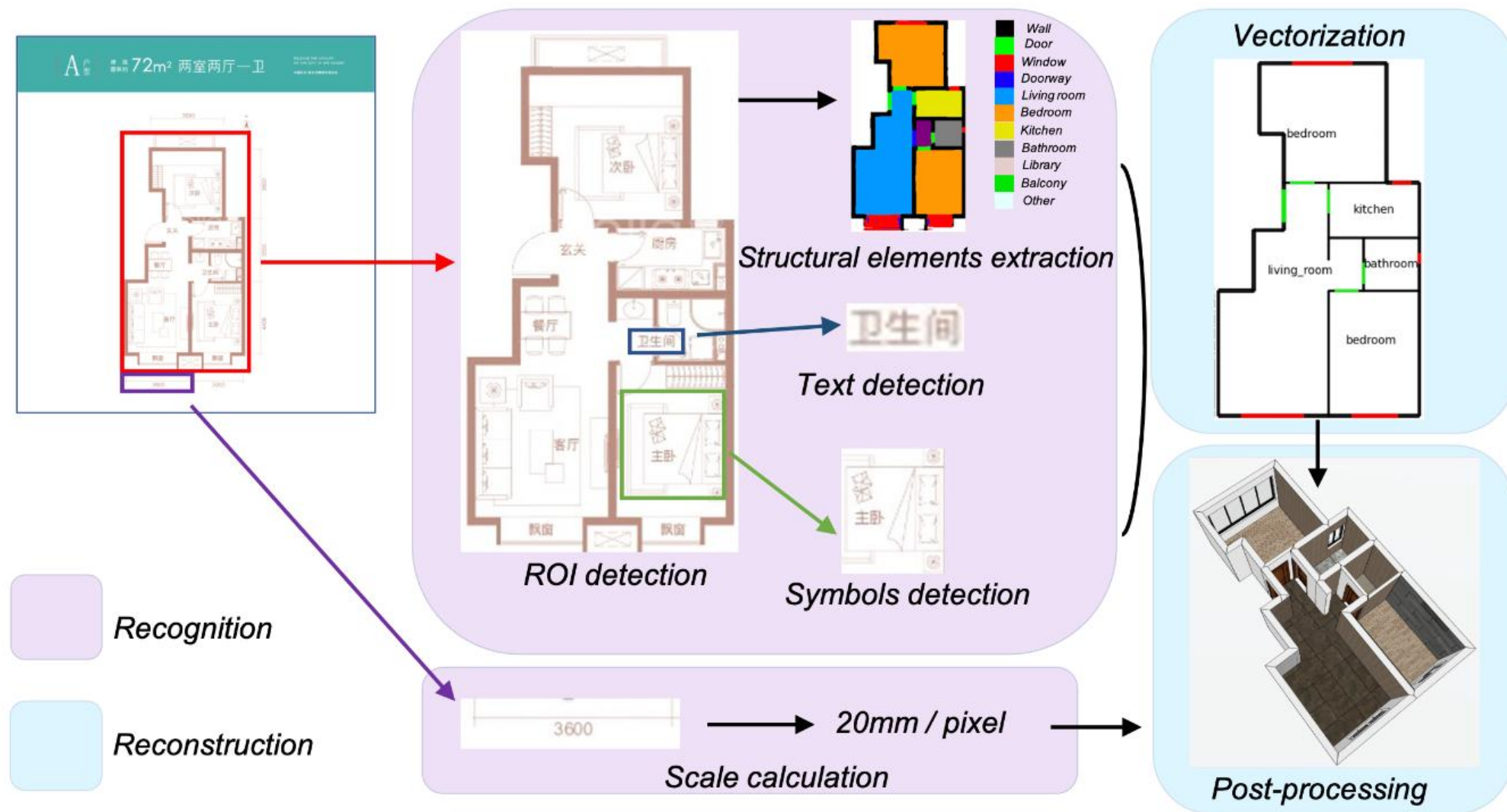
Jianyuan Wang, Minghao Chen, Nikita Karaev,
Andrea Vedaldi, Christian Rupprecht, David Novotny



Building Information Models for House Interiors



Residential floor plan recognition and reconstruction



Residential floor plan recognition and reconstruction



Controllable Makeup Transfer

(a) Shade-controllable Makeup Transfer



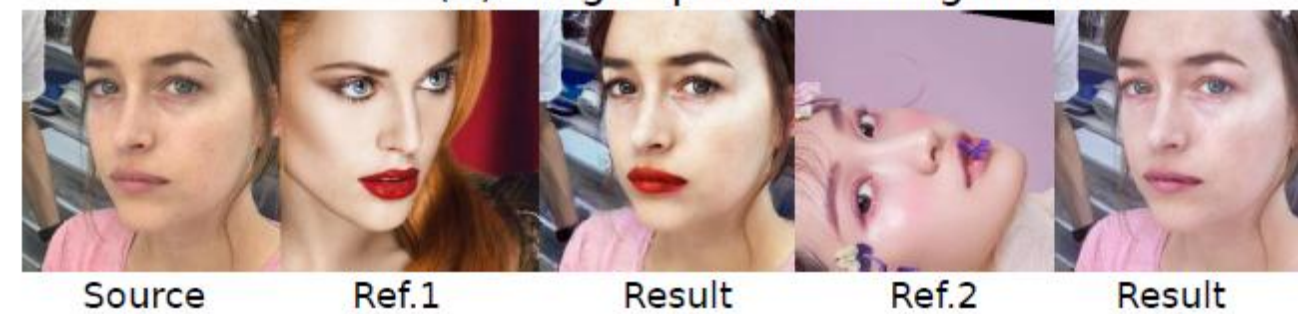
(b) Makeup Removal



(c) Part-specific Makeup Transfer



(d) Large spatial misalignment



Spatially-invariant Style-codes Controlled Makeup Transfer, CVPR'21

Beauty recommendation

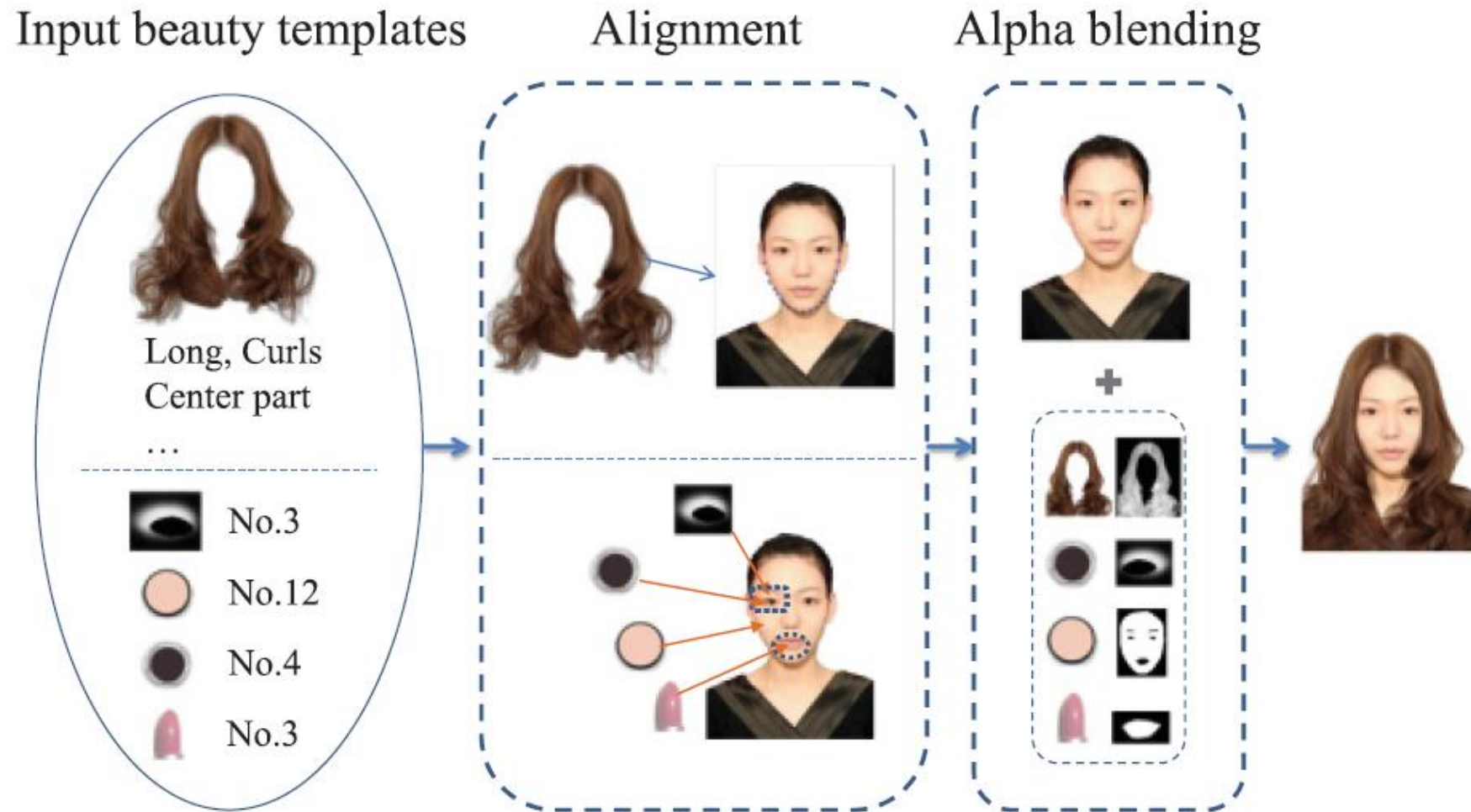
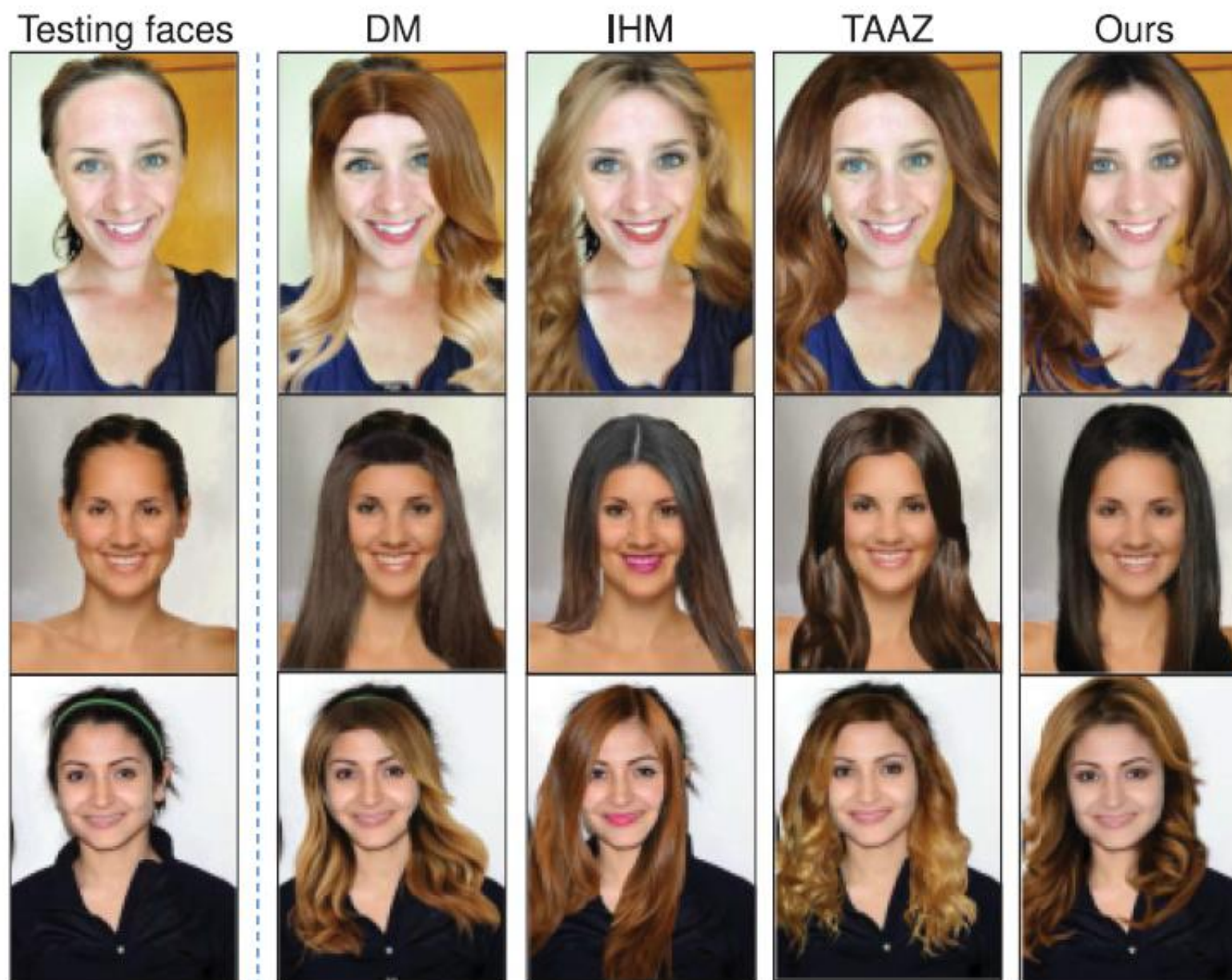
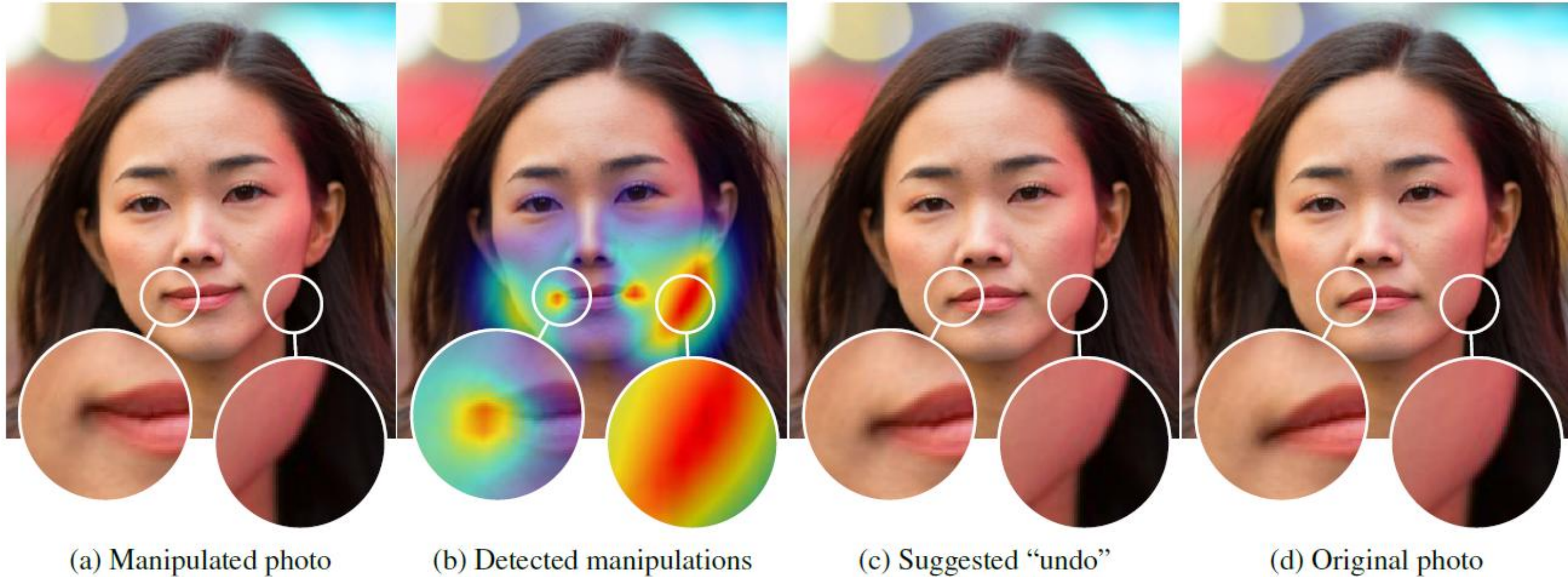


Fig. 6. The flowchat of the synthesis module.

Beauty recommendation



Detecting photoshopped faces



Detecting Photoshopped Faces by Scripting Photoshop, ICCV'19

Image synthesis



가상 피팅



Video/3D Generation

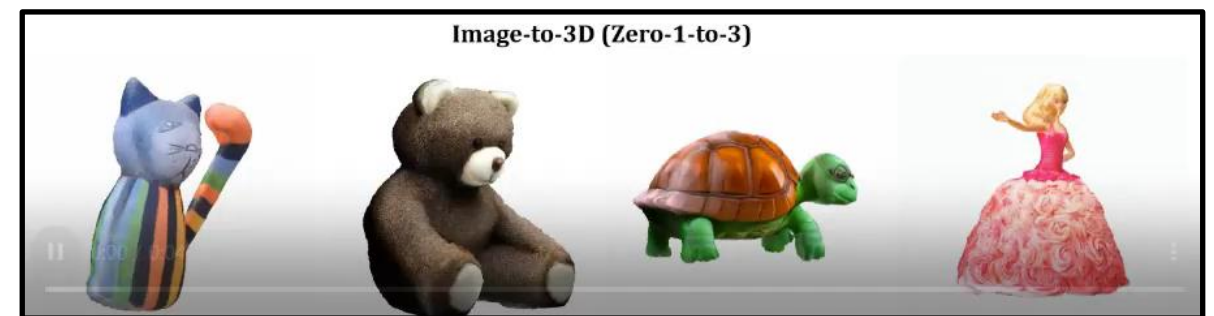
[Text-to-Video Generation]



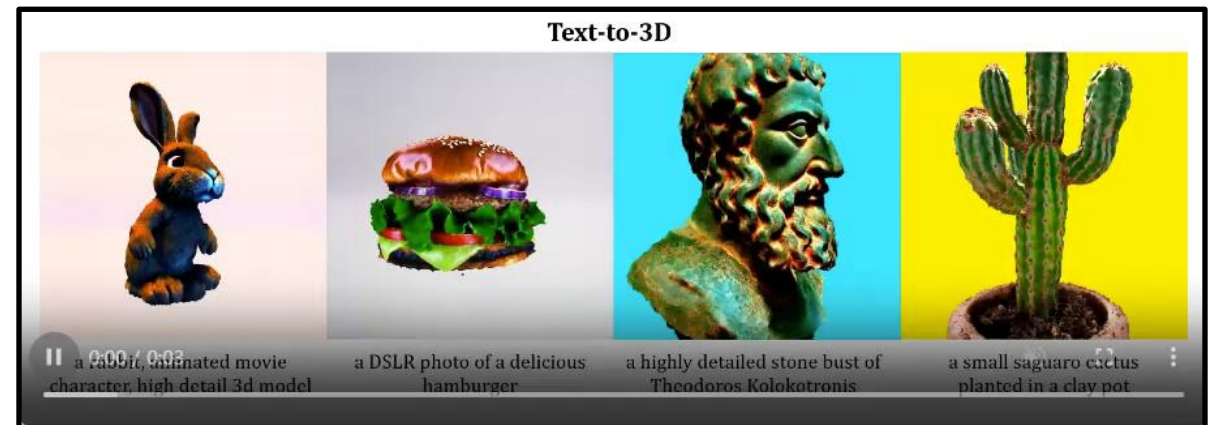
Prompt: Several giant wooly mammoths approach treading through a snowy meadow, their long wooly fur lightly blows in the wind as they walk, snow covered trees and dramatic snow capped mountains in the distance, mid afternoon light with wispy clouds and a sun high in the distance creates a warm glow, the low camera view is stunning capturing the large furry mammal with beautiful photography, depth of field.

Source : <https://openai.com/index/sora/>

[Image-to-3D Generation]



[Text-to-3D Generation]



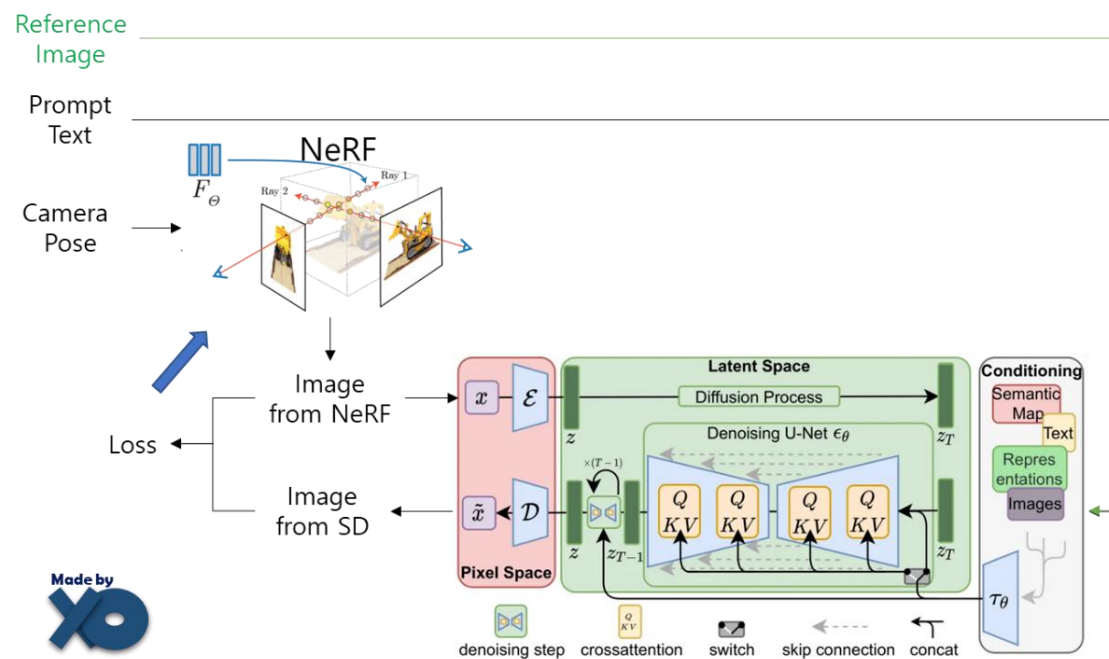
Source : <https://github.com/ashawkey/stable-dreamfusion>

Latent Diffusion Model (Stable Diffusion)

Rombach, Robin, et al. "High-resolution image synthesis with latent diffusion models." *CVPR*. 2022.

- Zero 1-to-3

Results



Latent Diffusion Model (Stable Diffusion)

Rombach, Robin, et al. "High-resolution image synthesis with latent diffusion models." *CVPR*. 2022.

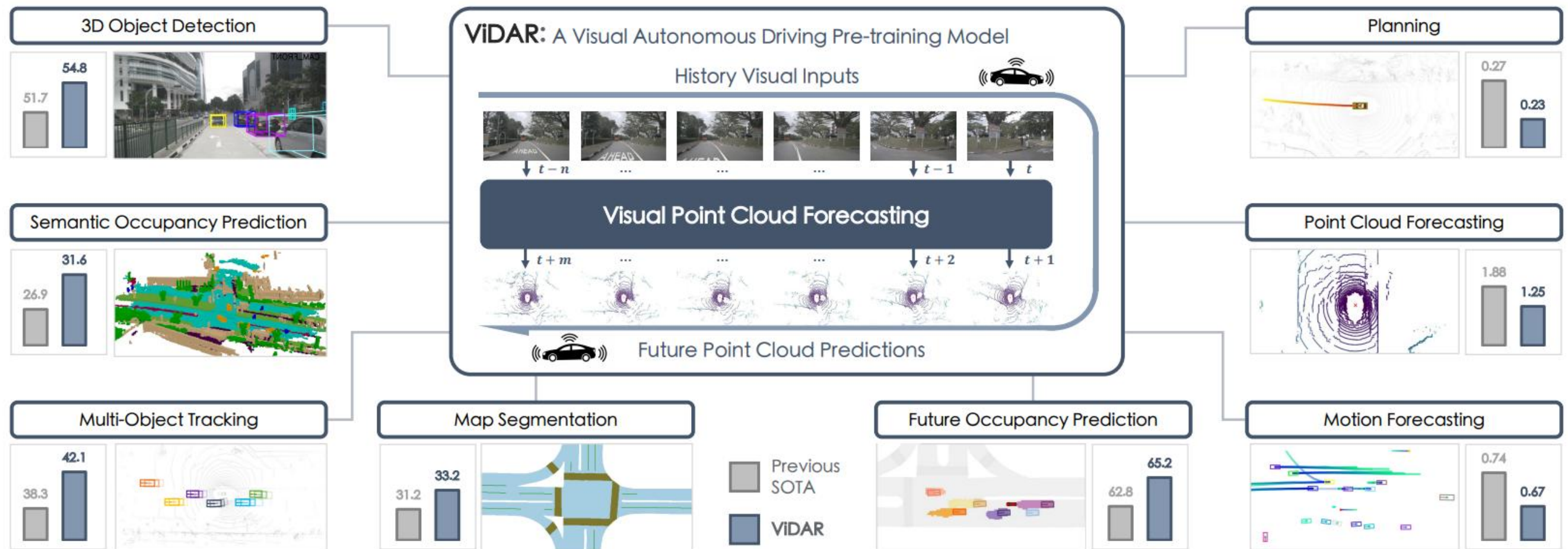
- VideoCrafter



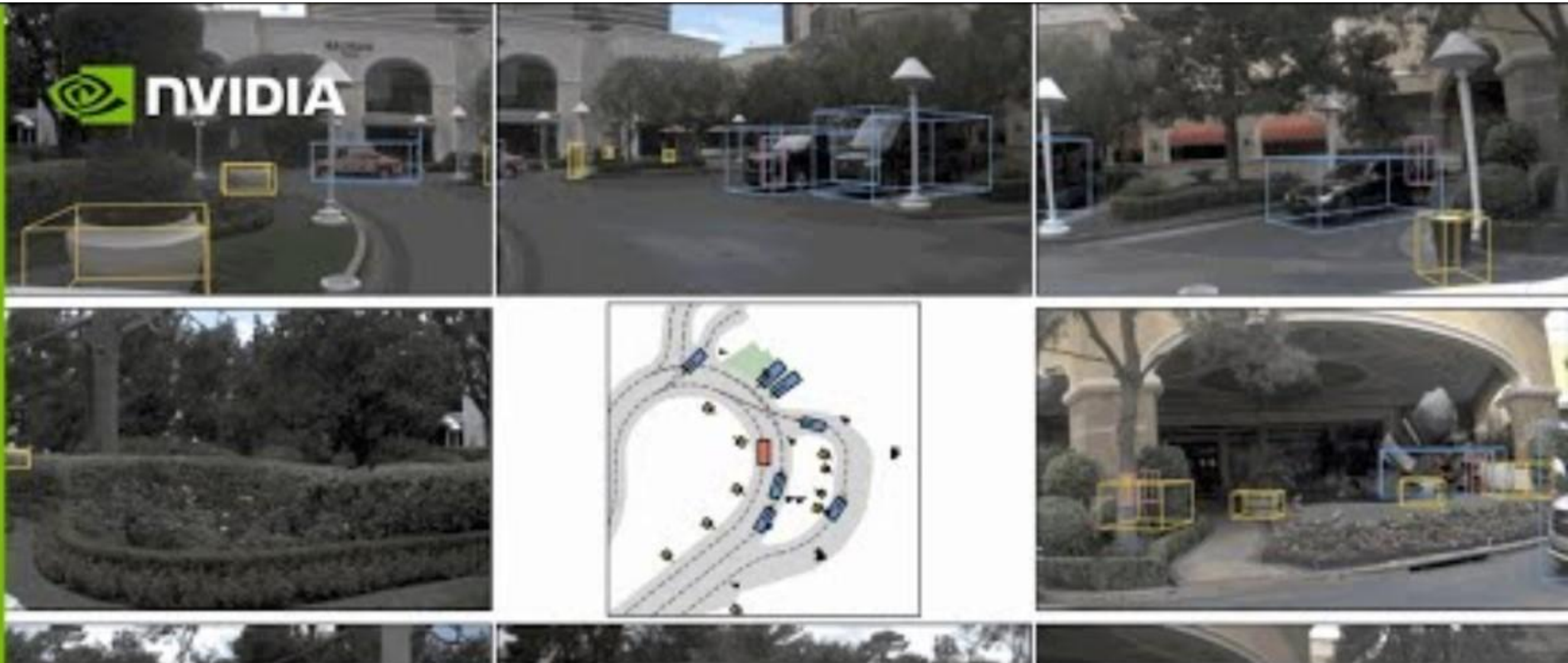
Results



Autonomous driving



Autonomous driving



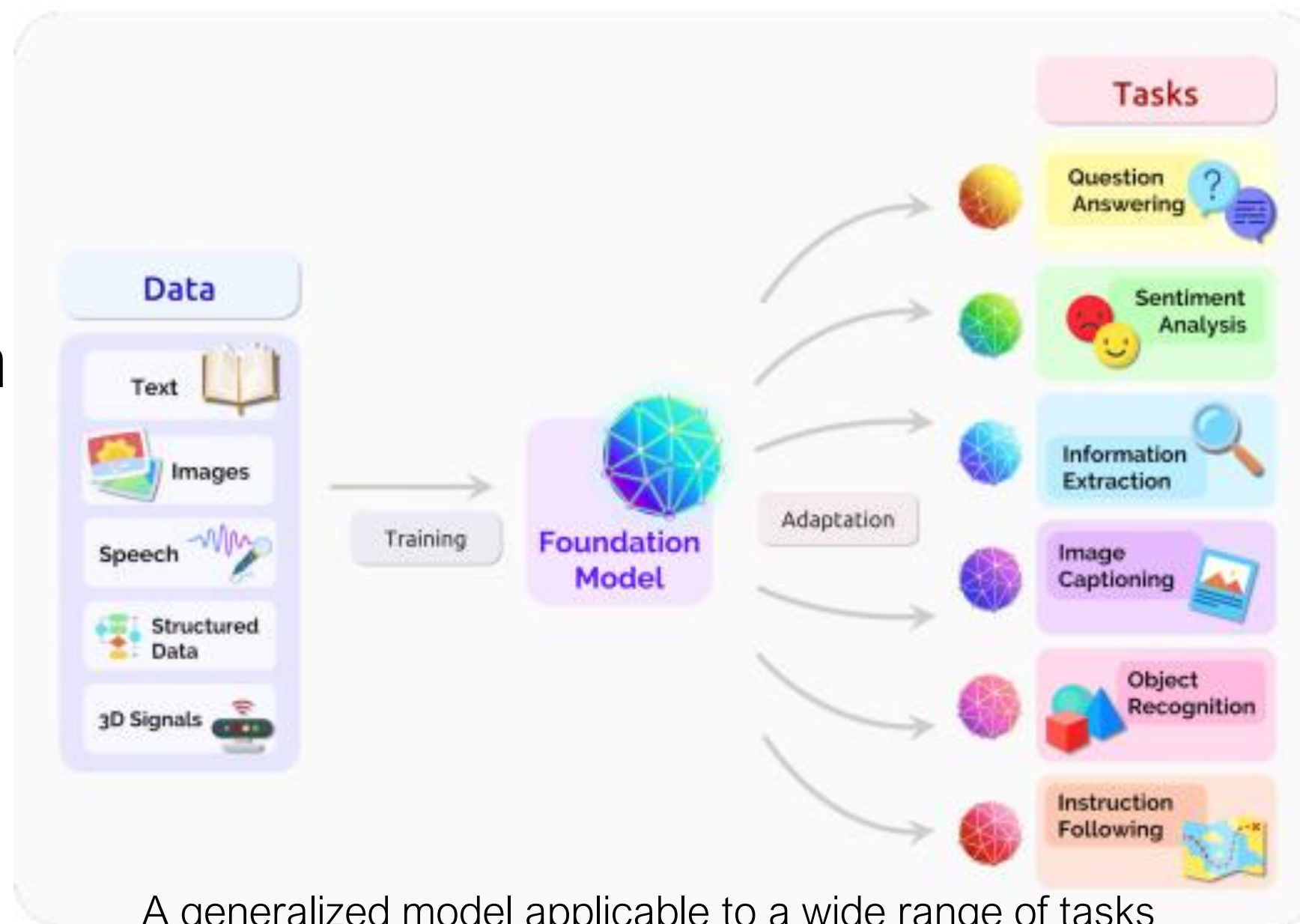
**End-to-End
Driving at Scale**

Foundation model

- What is a foundation model?

Foundation models are models built on **unlabeled data** using **self-supervision**

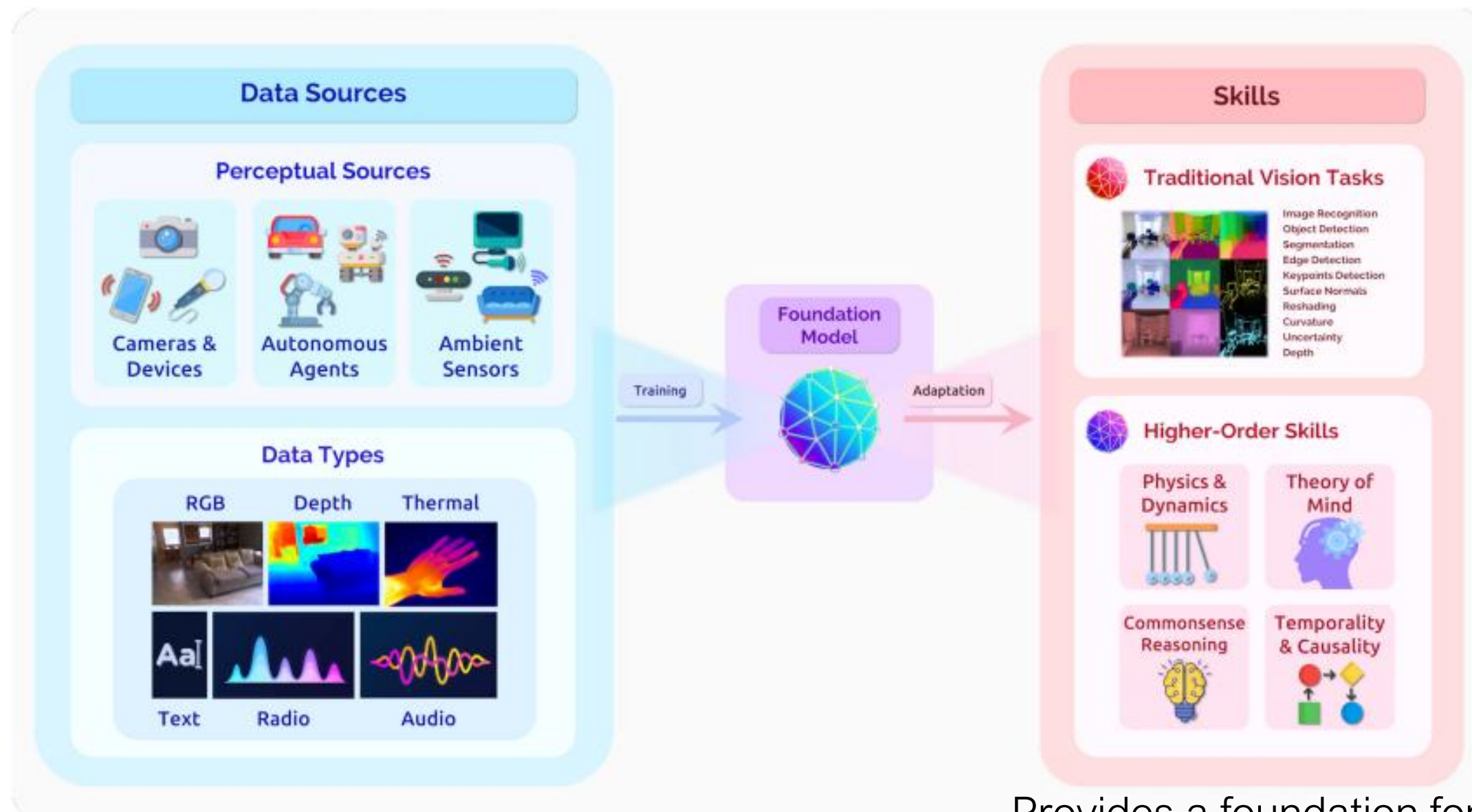
Trained on
extensive
data



A generalized model applicable to a wide range of tasks

Foundation model

- Generalization



Trained on diverse data sources without a specific purpose

Provides a foundation for creating new models specialized for specific domains

SAM (Segment Anything Model)

SAM 2

 Meta



Depth anything



<Perception to Description>

Image - Synthetic	Video - Synthetic	Video - Human	Open Source
- Documents - QA (PDFAcc, UCSF, 29.9M) - Charts - Caption, QA (ArxivCap, 1.8 M) - Natural Images - Captions (SA1B, Objects365, OpenImages, 15.3 M)	- Exocentric Videos - Captioning (YT-Temporal, 14 M) - MCQs (YT-Temporal, 3 M) - Egocentric Videos - Captions (Ego4D, 200 K) - QA (Ego4D, 700 K)	- Fine-grained - Activity QA (1.3 M) - How2 QA (1.4 M) - Spatio-Temporal - PLM-RTL0c (180 K) - PLM-RCap (180 K) - PLM-RDCap (117 K)	- Images - Single (4.4 M) - Multiple (293 K) - Videos - QA, MCQs (499 K) - Captions (315 K) - Grounding (81 K)
46.8M (Ours)	17.9M (Ours)	3.18M (Ours)	5.59M
Total 57.6M PLM Image and Video Data			5.59M Existing Data

Perception Language Model (PLM)

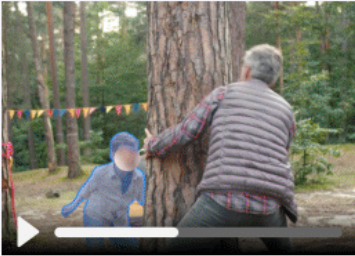
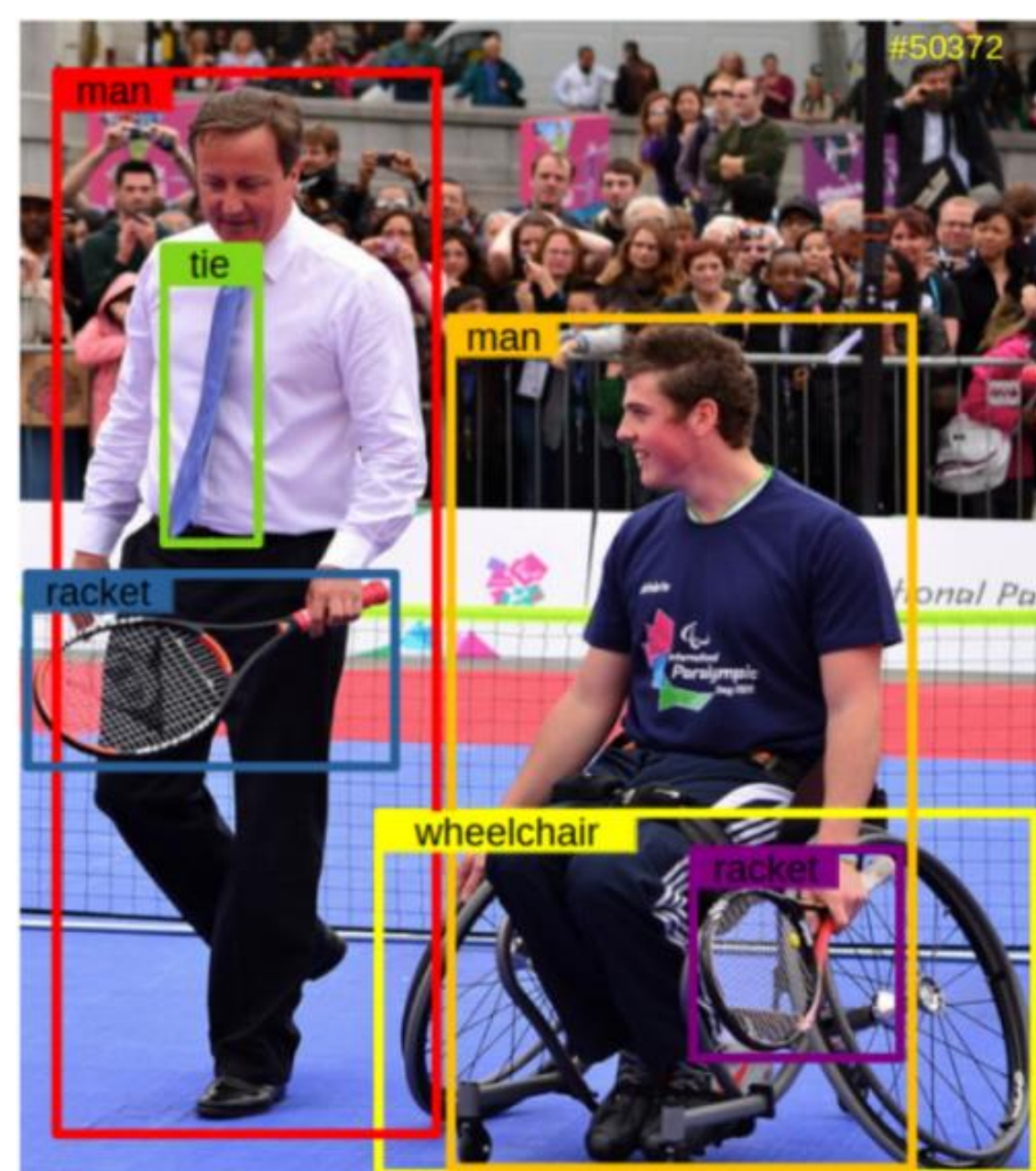
Image QA  Q: Which Place is shown in the image? A: A serene lake scene with a monument featuring a man on horseback.	Image OCR  Q: What location does this map represent? A: The map is of Kansai International Airport (KIX), which is located in Japan.	Multi Image  Q: Spot the difference between these two images. A: The color of the pole on the dog's right changes from black to light red.
Video QA  Q: What is the person doing in the video? A: Person is wearing ring in left hand and cutting the onions with a knife.	Video Cap.  Q: Briefly caption the video. A: The video shows a person making avocado sandwich by mashing the avocado and spreading it on bread.	PLM-FGQA  Q: How does a person pour the tea into the mug? A: The person moves the teapot up and down three times while pouring the tea into the mug.
PLM-RTL0c  Q: Localize the event "Person facing towards the camera is dancing." for the shaded region? A: [0, 13]	PLM-RCap  Q: Describe the highlighted subject in the video between frames [0, 13]. A: A person facing towards the camera is dancing.	PLM-RDCap  Q: Provide a dense caption for the shaded region. A: [0, 12] Kid is playing hide and seek and hides behind a tree. [13, 16] The kid reappears.

Image Captioning

- Involves automatically generating a natural language description for an image.



(a) image captioning

Output:

Men playing tennis on a tennis court

(b) dense captioning

Output:

Men playing a game of tennis

A man riding a wheelchair on a tennis court

A man standing on a tennis court

Men playing tennis on a tennis court

Men playing tennis on a court

People sitting on a bench

A man standing on a tennis court holding racket

(c) RefCap

Prompt:

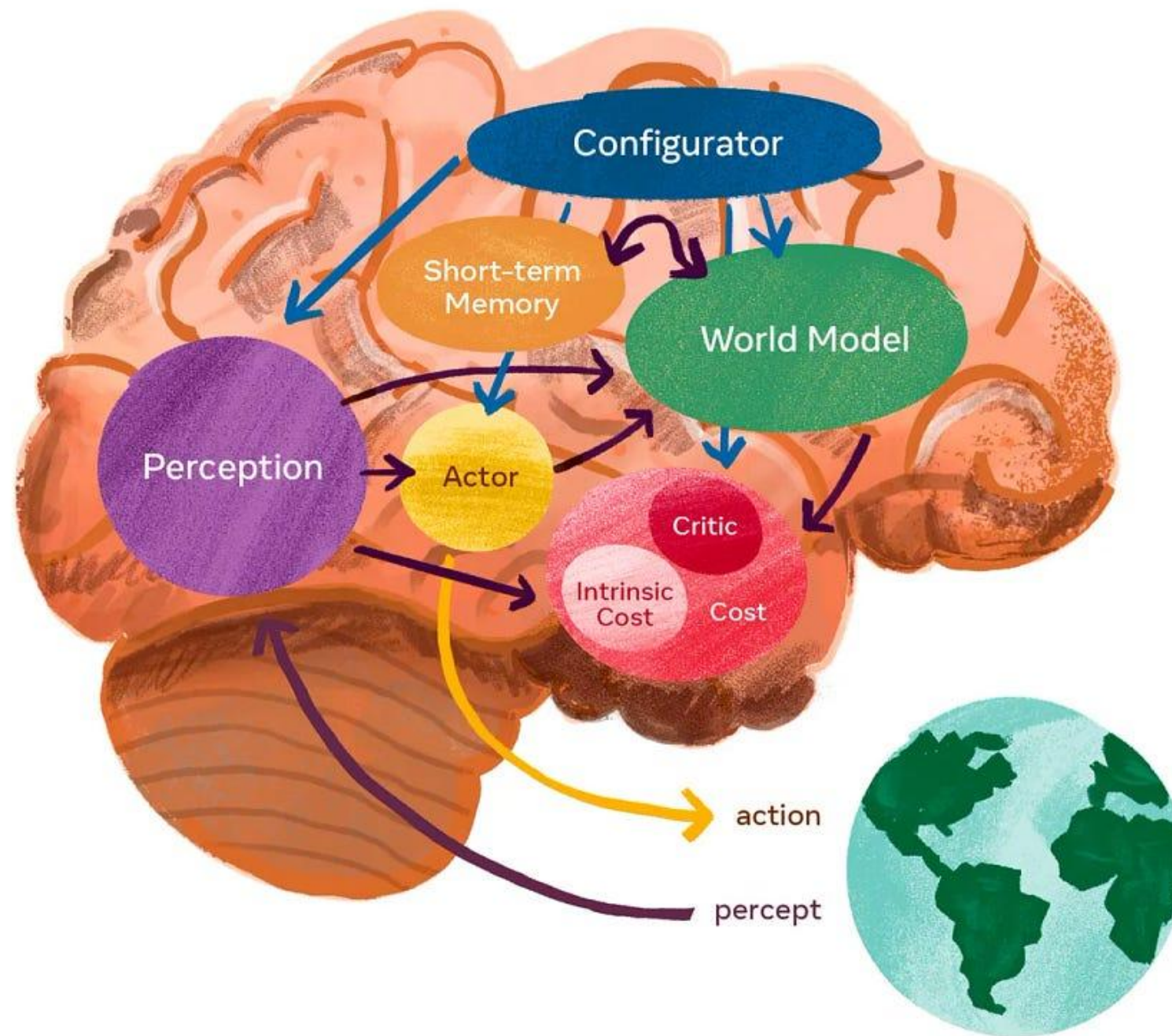
right man

Output:

A man riding a wheelchair with holding a racket

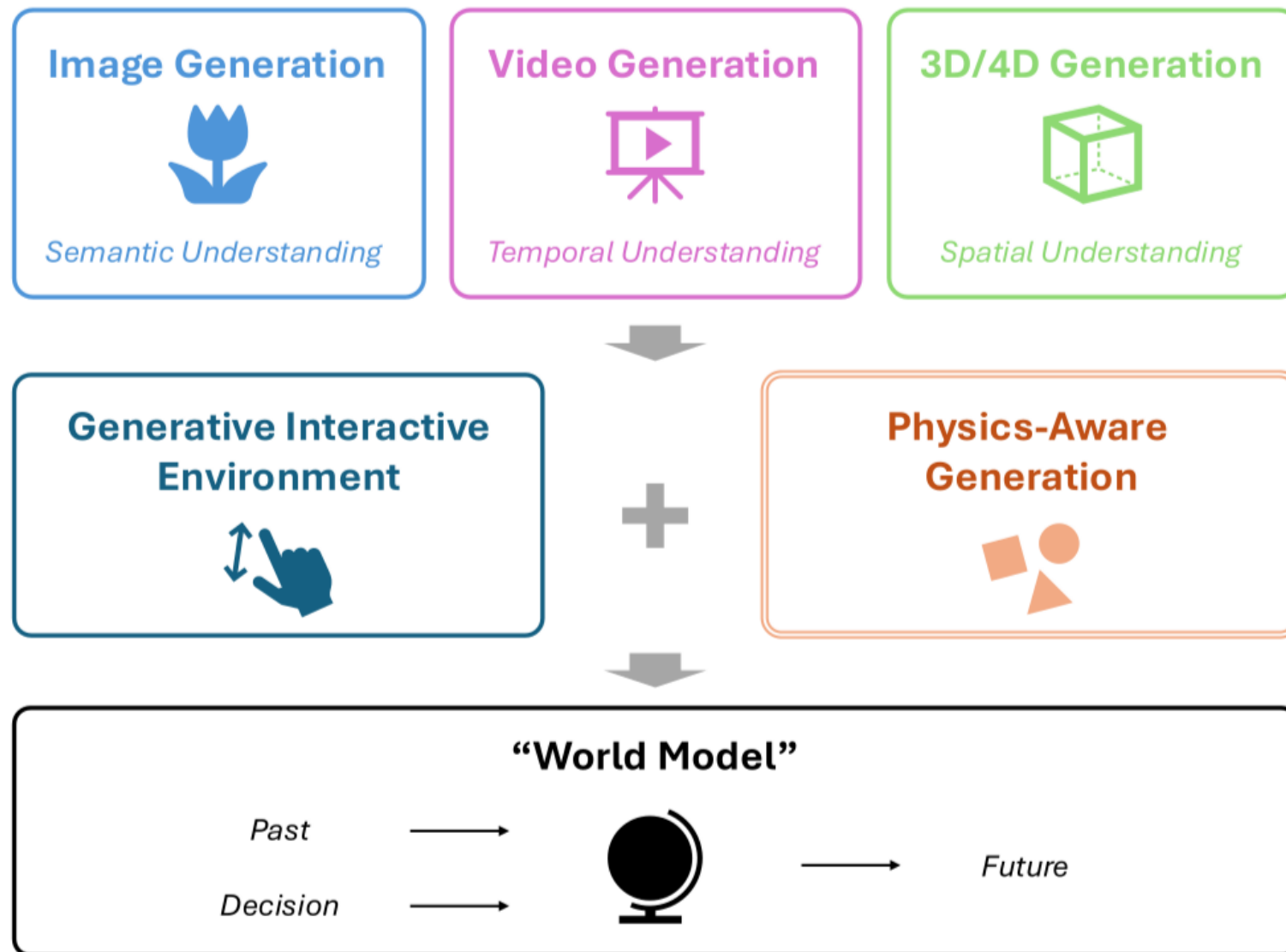
<Description to Depiction>





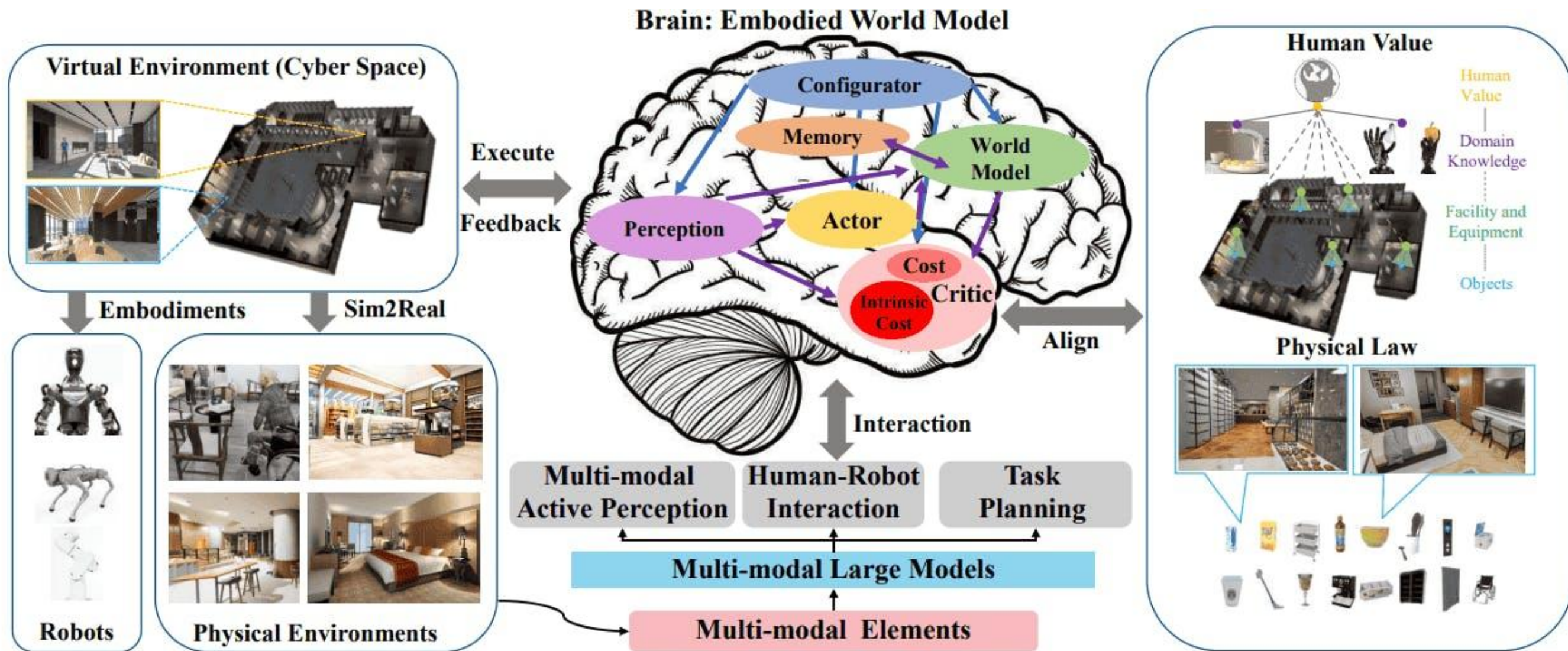
Physical AI

- A branch of AI that enables machines to perceive, understand, reason, and interact with the real, tangible world, rather than operating solely in a digital domain



Physical AI

- A branch of AI that enables machines to perceive, understand, reason, and interact with the real, tangible world, rather than operating solely in a digital domain





VLA (Vision Language Action Model)

- Class of multimodal foundation models that integrates vision, language and actions.

(actions are expressed as a text tokens.)

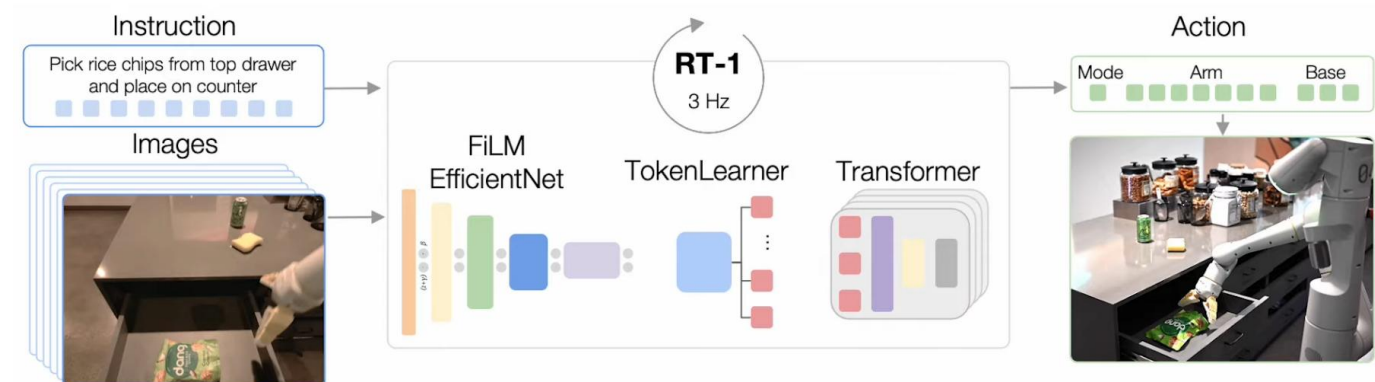
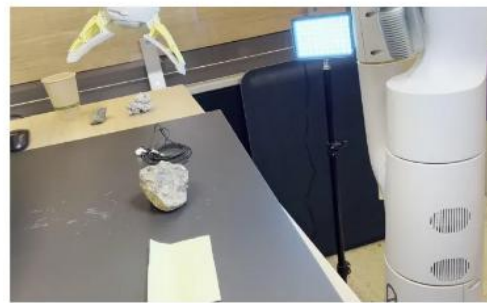
- VLAs are generally constructed by fine-tuning a vision-language model (VLM).

Instruction:

I need to hammer a nail,
what object from the scene
might be useful?

Prediction:

Rocks. Action: 1 129 138 122
132 132 106 127



"Operating system of the physical AI world"



XPENG VLA 2.0

Higher Efficiency & Faster Response

 XPENG

Guangzhou, China, November 5, 2025 - XPENG VLA 2.0.
<https://youtu.be/8qsLreyJ1qQ?si=eQJaCwSefCCq8bFW>

World Foundation Model

- The World Model is a Generative AI model that understands the dynamics of the real world, including physical and spatial properties.
 - The model generates visual content using input data including text, images, videos, and movements.
- It also understands the physical properties of real environments by learning to express and predict dynamics such as movement, force, and spatial relationships through sensing data.



Omniverse (Simulator)

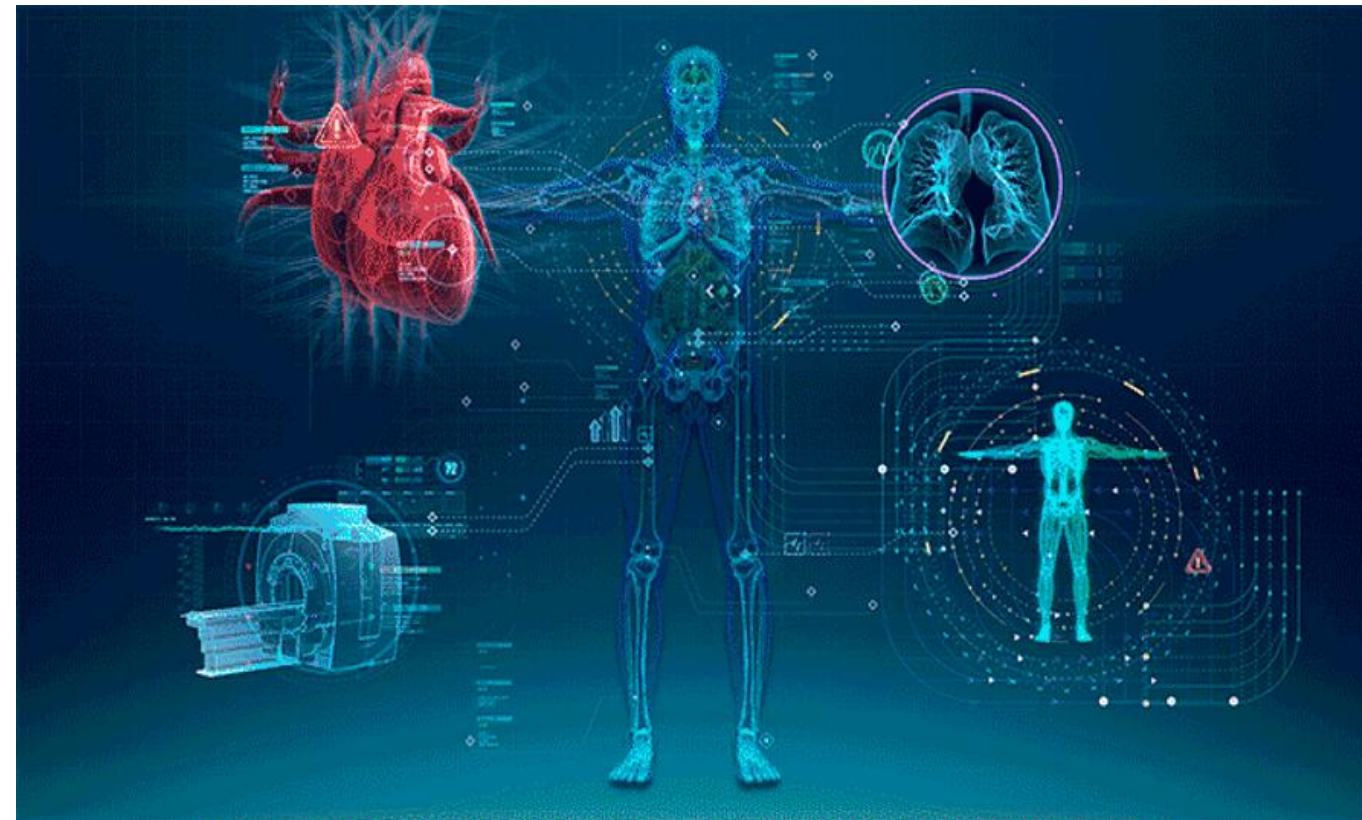
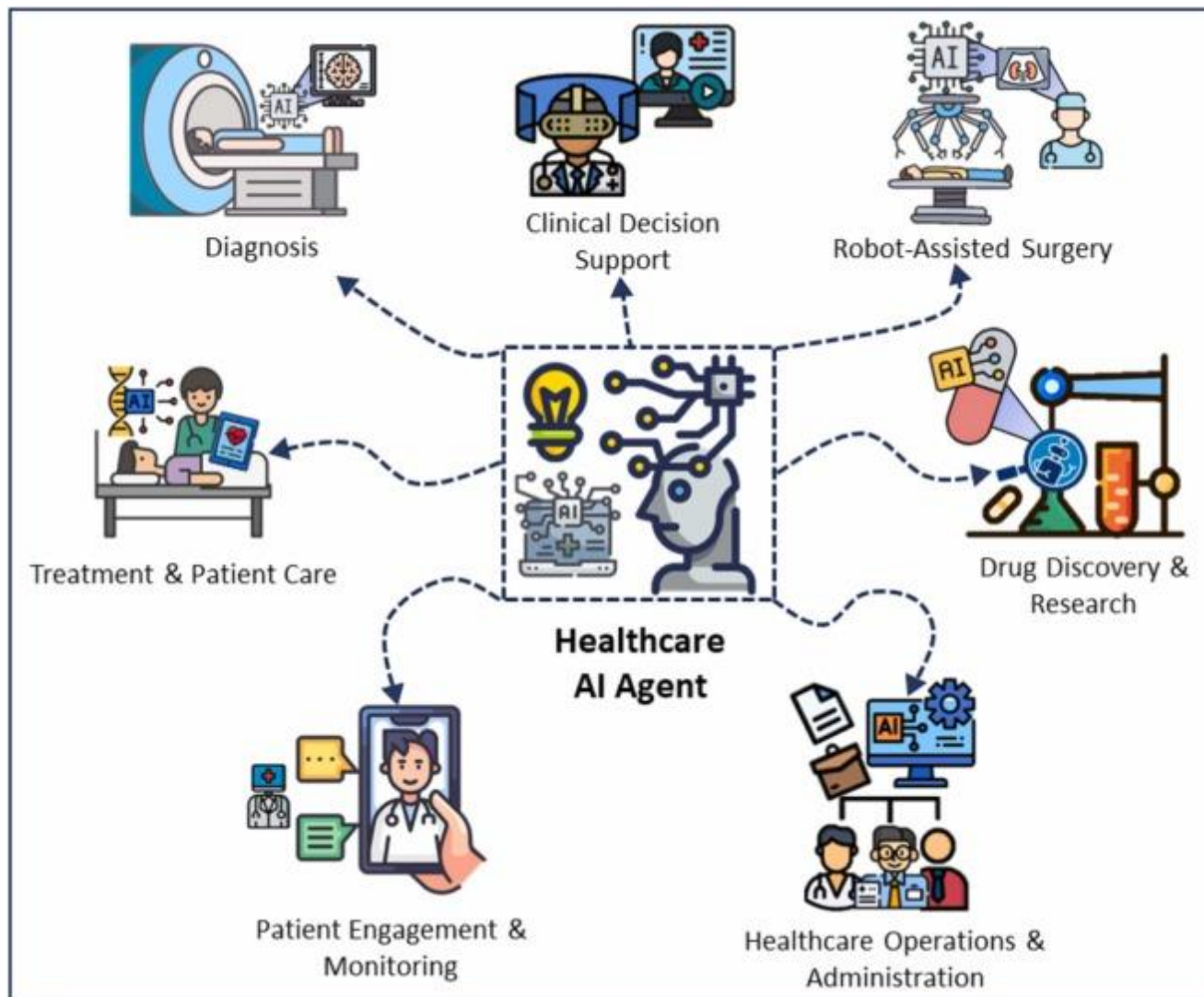


Cosmos (World Generation)

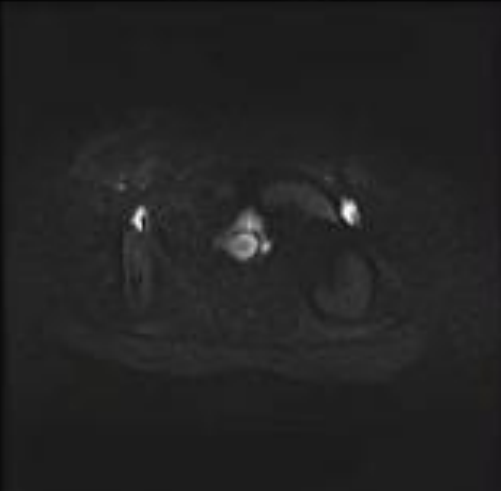
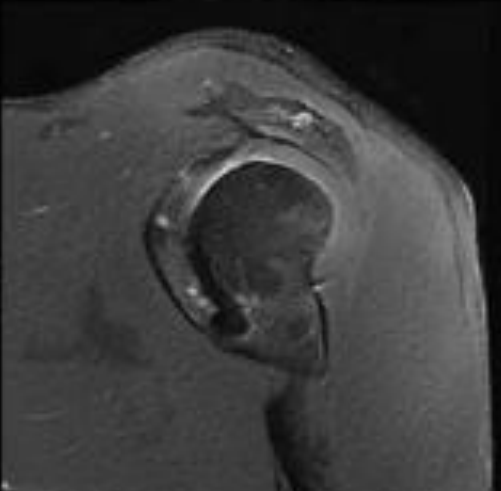
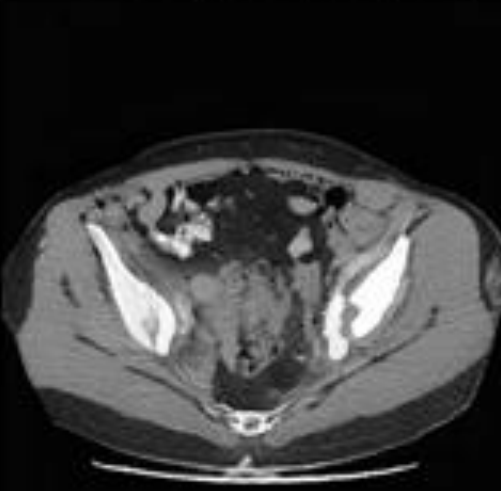
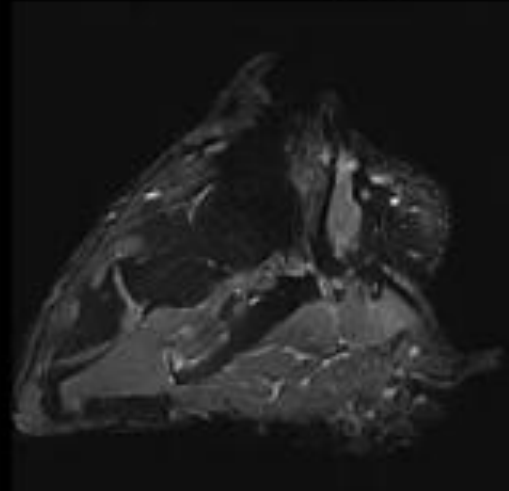
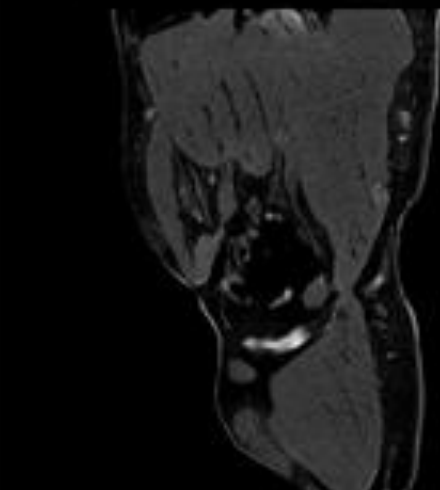
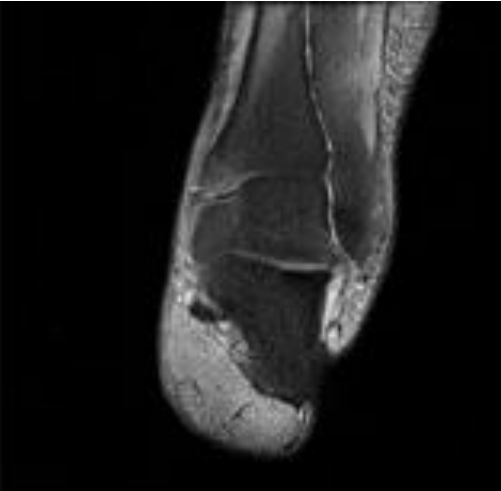
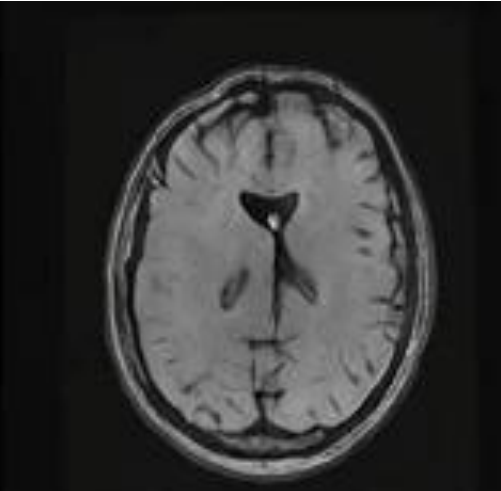
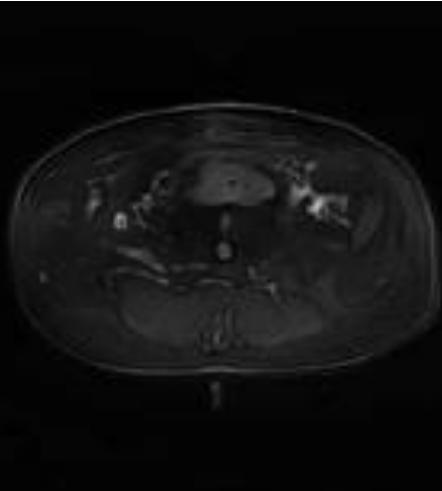


Medical AI

- Medical AI enables robots to interpret medical data and support diagnosis, treatment, patient monitoring, and clinical workflows.







BiomedCLIP: image-to-text retrieval results

B

Caption

Chest X-ray on admission showing a large pleural effusion on the right.

Caption

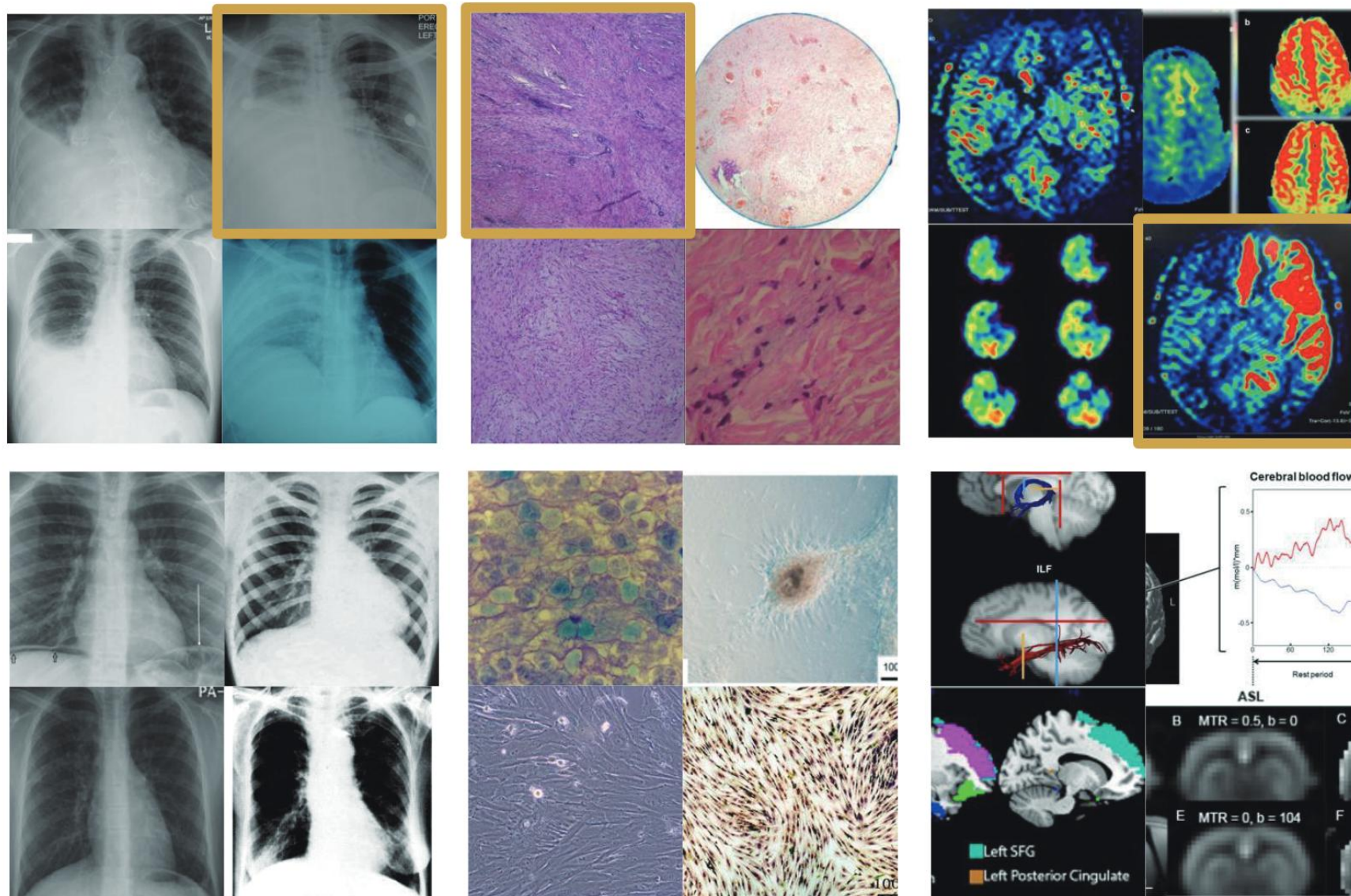
Photomicrograph showing proliferation of spindle cells with wavy nuclei in collagenous to myxoid stroma (Hematoxylin and Eosin staining, 40×).

Caption

ASL image shows increased cerebral blood flow (CBF) in left fronto-parieto-temporal region.

BiomedCLIP

CLIP

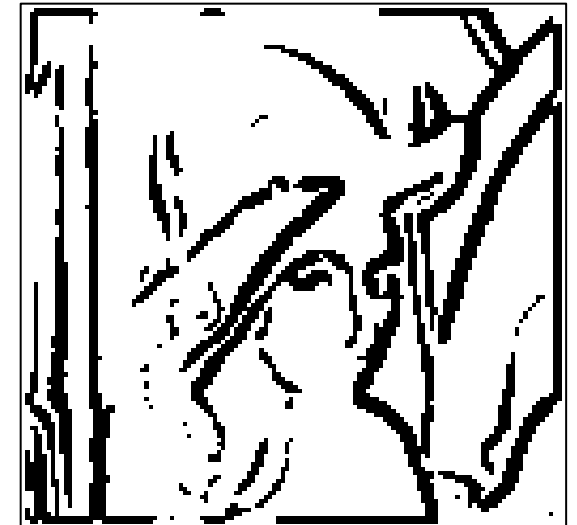


Course logistics

Topics to be covered

Image processing:

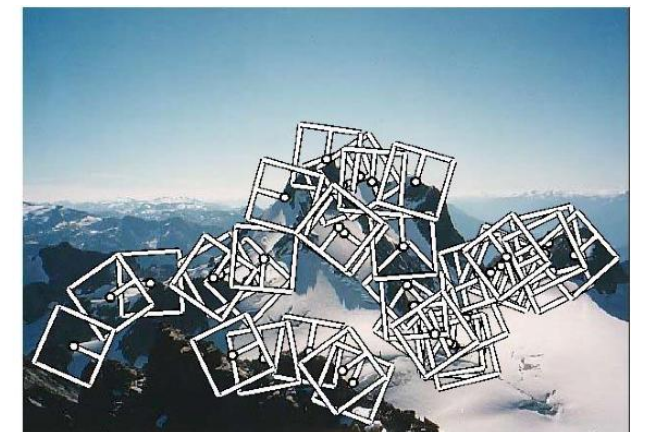
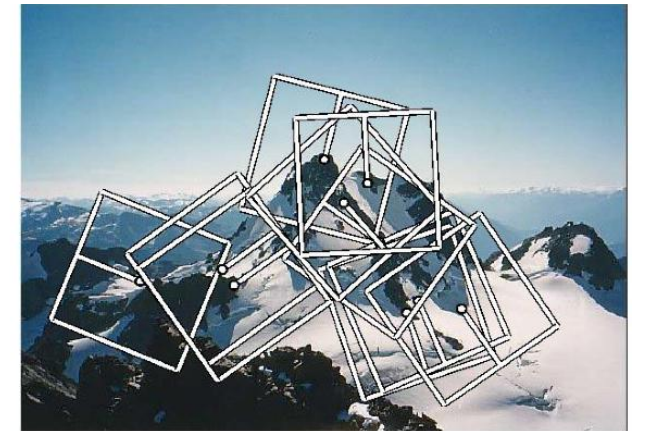
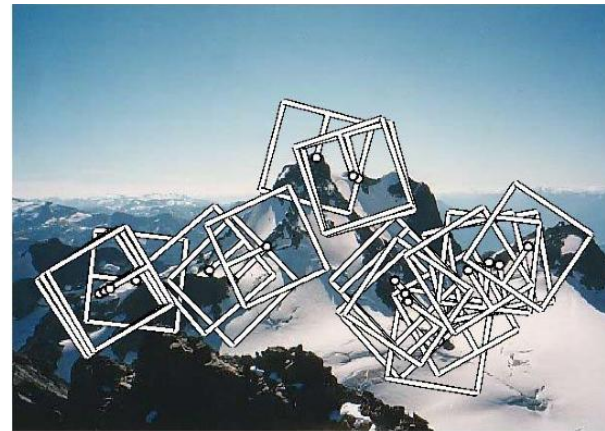
- Basics of filtering.
- Image pyramids.
- Gradients and lines.
- Hough transforms.



Topics to be covered

Feature detection and correspondences:

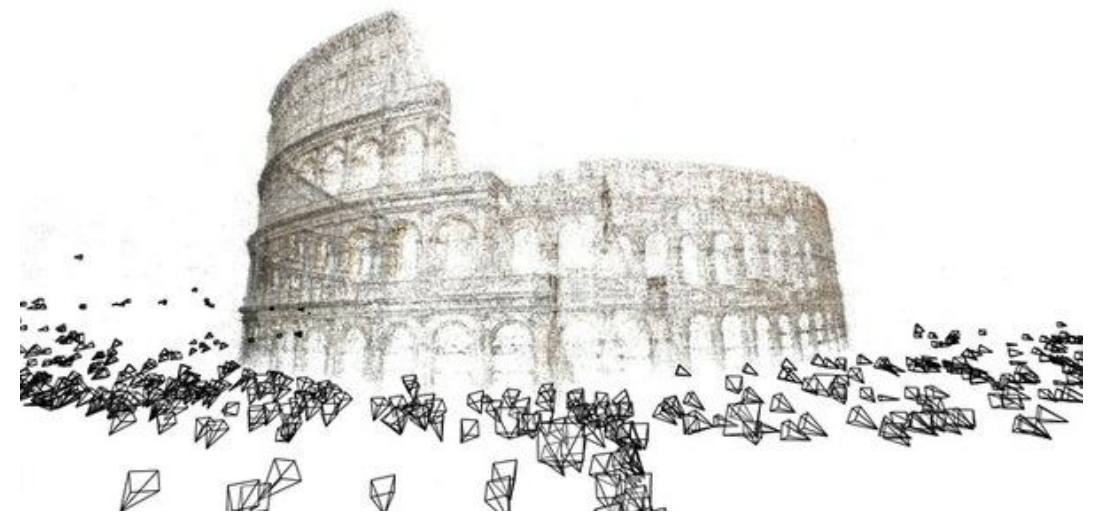
- Corner detection.
- SIFT et al.
- Feature descriptors.
- RANSAC.



Topics to be covered

Transformations and geometry:

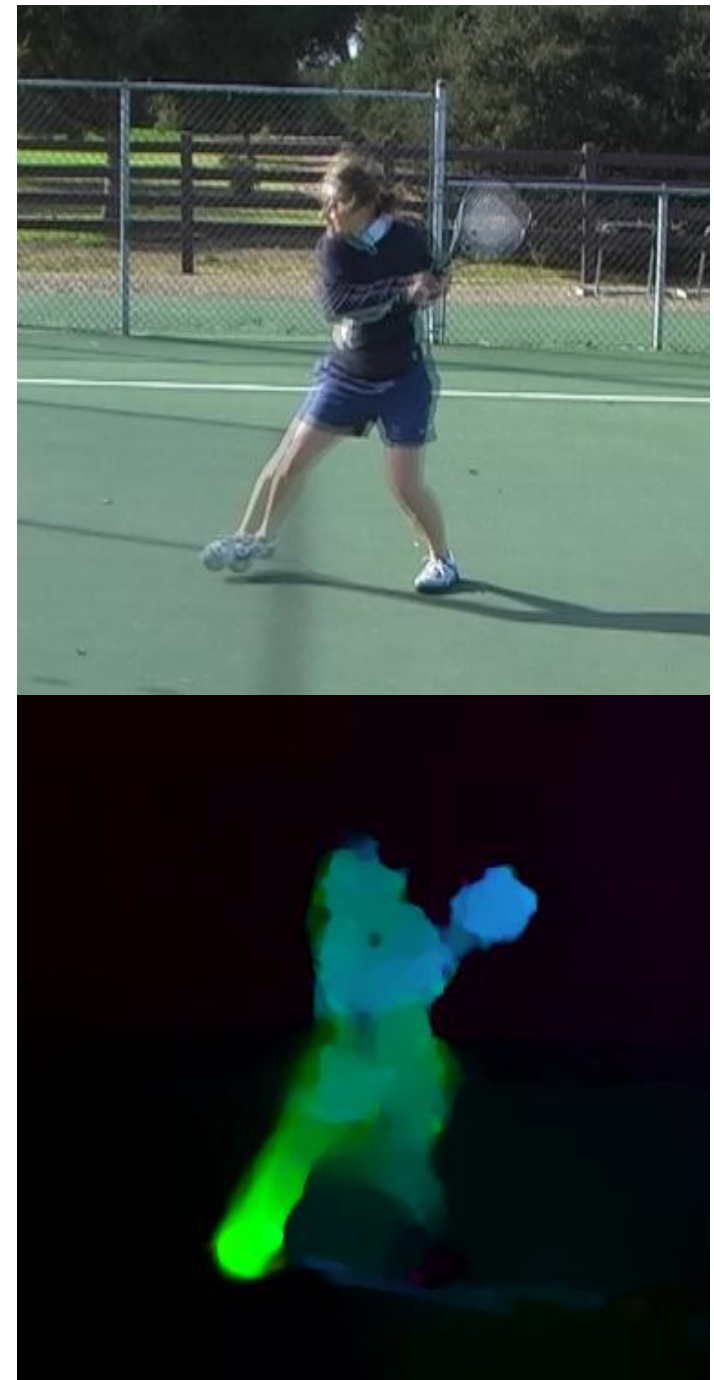
- Homographies and image alignment.
- Camera models.
- Fundamental matrix.
- Epipolar geometry and stereo.
- Structure from motion.



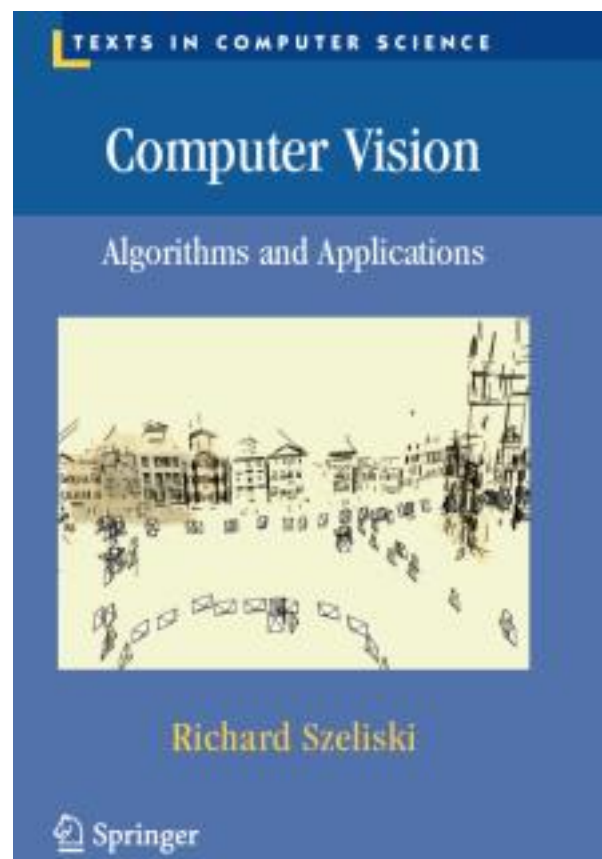
Topics to be covered

Dealing with motion:

- Optical flow (LK, HS).
- Image registration.
- Kalman Filtering.
- Tracking (KLT, Mean-Shift).



Reference

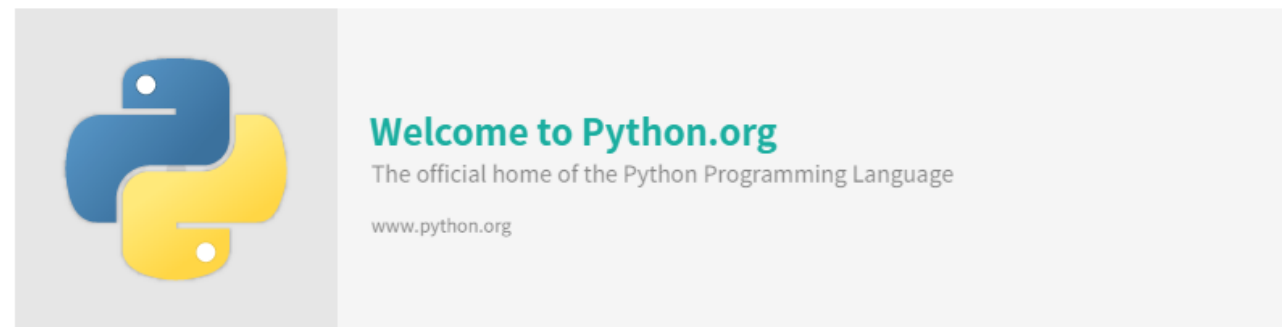


PDF online

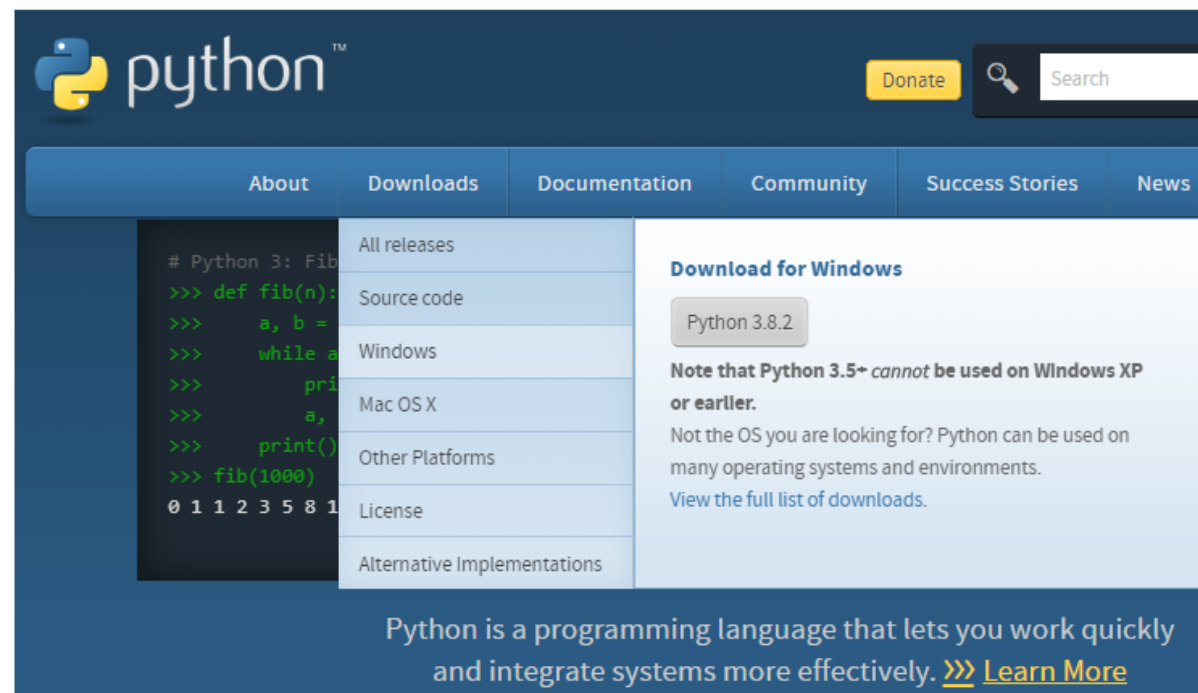
<http://szeliski.org/Book/>

Python, OpenCV 설치

<https://www.python.org/>



- 링크를 통해 파이썬 공식 홈페이지로 들어갑니다.



파이썬 공식 홈페이지

Python, OpenCV 설치

- Downloads - Windows 클릭

- **Python 3.6.8** - Dec. 24, 2018
Note that **Python 3.6.8** *cannot* be used on Windows XP or earlier.
 - Download [Windows help file](#)
 - Download [Windows x86-64 embeddable zip file](#)
 - Download [Windows x86-64 executable installer](#)
 - Download [Windows x86-64 web-based installer](#)
 - Download [Windows x86 embeddable zip file](#)
 - Download [Windows x86 executable installer](#)
 - Download [Windows x86 web-based installer](#)

- Python 3.6.8 을 찾아 **Download Windows x86-64 executable installer** 클릭

- exe 파일 실행

Python, OpenCV 설치



- Add Python 3.6 to PATH 체크박스 체크
- Customize installation 클릭

Python, OpenCV 설치

```
C:\Users\USER>cd C:\Python36 #cd는 change directory의 약자입니다. (이거 안해도 되긴합니다.)
```

```
1 C:\Python36>python --version
```

```
2 Python 3.6.8
```

- `pip install opencv-contrib-python==3.3.1.11`